21-701: Discrete Mathematics

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For the latest version of these notes, visit

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Chapter 1

Probability

Almost every argument in these notes produces an object by showing that a random one works, so the vocabulary for talking about random objects had better come first. This chapter collects it. Very little of what follows is new—probability spaces, expectation, variance—and none of it is proved at length; what matters is that the two inequalities the rest of the course uses by name, Markov's and Chebyshev's, are written down somewhere rather than assumed. Conditional expectation gets a little more care than the rest, because Chapter 4 uses it heavily. The last section sets down the asymptotic notation—o, ω, O and Ω—that the later chapters use whenever something grows with n.

The course itself begins in Chapter 2, and a reader who would rather see a graph than a sigma-algebra can start there and come back when a probability turns up.

1.1 Probability Spaces

Almost everything in these notes is probabilistic, so it is worth setting down what we are going to use before we use it. None of this section goes beyond a first course, and we will take the measure theory on trust.

Recall the following: if $(\Omega, \mathcal{F}, \Pr)$ is a probability space, this means

- Ω is a set of outcomes, called a **sample space**;

- $\mathcal{F}$ is a sigma-algebra on Ω;
- $\Pr : \mathcal{F} \to [0, \infty)$ is a measure that only takes values in $[0, 1]$, with $\Pr(\Omega) = 1$.

An **event** is a measurable set (ie, an element of $\mathcal{F}$), and a random variable is a measurable $\Omega \to \mathbb{R}$ function.

The one notion from this review that the rest of the notes use constantly is independence, in three forms.

Definition 1.1.1 (Independence)**.** Two events A and B are **independent** if $\Pr(A \cap B) = \Pr(A) \cdot \Pr(B)$. Random variables $X_1, \ldots, X_n$ are **independent** if for all Borel sets $A_1, \ldots, A_n \subseteq \mathbb{R}$ the events $X_1^{-1}[A_1], \ldots, X_n^{-1}[A_n]$ (all of which lie in $\mathcal{F}$ because the X_i are measurable) satisfy $\Pr\left(X_1^{-1}[A_1] \cap \cdots \cap X_n^{-1}[A_n]\right) = \Pr\left(X_1^{-1}[A_1]\right) \cdots \Pr(X_n^{-1}[A_n])$, and an infinite family is independent if every finite subfamily is. An event A is **independent of a family of events** $B_1, \ldots, B_n$ if $\Pr(A \cap C) = \Pr(A) \cdot \Pr(C)$ for every event C obtained from $B_1, \ldots, B_n$ by intersections, unions and complements.

The third form is stronger than asking A to be independent of each B_i separately: if ξ_1, ξ_2 are independent fair bits and $\xi_3 = 1 \oplus \xi_1 \oplus \xi_2$, then $\{\xi_3 = 1\}$ is independent of each of $\{\xi_1 = 1\}$ and $\{\xi_2 = 1\}$, but not of their intersection. It is what the Lovász local lemma (Theorem 5.1.2) needs.

In this course, Ω will usually be finite, $\mathcal{F}$ will usually be the $\mathcal{P}(\Omega)$, and all functions $\Omega \to \mathbb{R}$ will be random variables.

1.2 Expectation

The expectation is the integral of a random variable over the probability space with respect to the probability measure.

Two things stand out about independent random variables.

1. If $\eta_1, \eta_2, \eta_3, \ldots$ are independent and $f, g, h, \ldots$ are measurable functions, then $f(\eta_1), g(\eta_2), h(\eta_3, \eta_4), f($
   are all independent.

2. If $\eta_1, \ldots, \eta_n$ are independent, then

$$\mathbb{E}\left(\prod_{i=1}^{n} \eta_i\right) = \prod_{i=1}^{n} \mathbb{E}(\eta_i)$$

1.2.1 Conditional Expectation

Often we learn something partway through and want the expectation of what is left. That is an expectation again, taken over a smaller sample space.

Given an event A with $\Pr(A) \neq 0$ and a random variable X, we can define

$$\mathbb{E}(X \mid A) = \frac{1}{\Pr(A)} \sum_{x \in A} X(x) \cdot p(x)$$

where $p(x)$ is the density function. It is easy to show that

$$\mathbb{E}(X \mid A) = \sum_{r \in X[\Omega]} r \cdot \Pr(X = r \mid A)$$

Given *two* random variables X and Y, we denote by $\mathbb{E}(X \mid Y)$ the **random variable**

$$\omega \mapsto \mathbb{E}(X \mid Y = Y(\omega))$$

The point of packaging it as a random variable rather than as a list of numbers is that it can then be averaged again.

Theorem 1.2.1 (Tower Property of Conditional Expectation). *For all random variables X and Y,*

$$\mathbb{E}(\mathbb{E}(X \mid Y)) = \mathbb{E}(X)$$

The above type-checks because $\mathbb{E}(X \mid Y)$ is a random variable.

Several random variables at once are no different: $\mathbb{E}(X \mid Y_1, \ldots, Y_k)$ is the random variable $\omega \mapsto \mathbb{E}(X \mid Y_1 = Y_1(\omega), \ldots, Y_k = Y_k(\omega))$, conditioning on the one event that records what all of them did. Both this and $\mathbb{E}(X \mid Y)$ need each of those events to have nonzero probability, which on a finite Ω whose outcomes all have positive probability—the only case we use—every ω supplies.

Section 4.3 uses all of this rather more heavily than anything before it does.

1.3 Deviation from the Mean

Knowing what a random variable averages to is not the same as knowing what it does, and most of what we want from a random object is of the second kind. Two inequalities get us most of the way, and the whole of Chapter 4 is what to do when they do not.

The first says that a nonnegative random variable cannot often be much larger than its mean.

Theorem 1.3.1 (Markov's Inequality). *Let X be a nonnegative random variable and let $a > 0$. Then,*

$$\Pr(X \geq a) \leq \frac{\mathbb{E}(X)}{a}$$

Proof. Let A be the event that $X \geq a$. Because X is nonnegative, $X \geq a\mathbf{1}_A$ everywhere on Ω: off A the right-hand side is 0, and on A it is a, which is at most X. Taking expectations gives $\mathbb{E}(X) \geq a \Pr(A)$. □

Applied to X itself this is usually all we want, but applied to some other nonnegative random variable built out of X it is much sharper. The standard choice is the squared deviation.

Definition 1.3.2 (Variance). The **variance** of a random variable X is

$$\operatorname{Var}(X) = \mathbb{E}\Big((X - \mathbb{E}(X))^2\Big) = \mathbb{E}\big(X^2\big) - \mathbb{E}(X)^2$$

Variance is not linear, but it is additive on independent random variables: if $X_1, \ldots, X_n$ are independent then $\operatorname{Var}(\sum_i X_i) = \sum_i \operatorname{Var}(X_i)$, because every cross term $\mathbb{E}(X_i X_j) - \mathbb{E}(X_i)\,\mathbb{E}(X_j)$ with $i \neq j$ vanishes.

Feeding the squared deviation into Theorem 1.3.1 gives Chebyshev's inequality.

Theorem 1.3.3 (Chebyshev's Inequality). *Let X be a random variable and let $a > 0$. Then,*

$$\Pr(|X - \mathbb{E}(X)| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Proof. The random variable $Y = (X - \mathbb{E}(X))^2$ is nonnegative and has $\mathbb{E}(Y) = \mathrm{Var}(X)$, and $|X - \mathbb{E}(X)| \geq a$ exactly when $Y \geq a^2$. So Theorem 1.3.1, applied to Y at a^2, gives what we want. □

The reason this is worth having is that it says something the expectation cannot. Knowing $\mathbb{E}(X)$ is large tells us nothing about whether X is ever nonzero; knowing in addition that X does not stray far from $\mathbb{E}(X)$ does.

Corollary 1.3.4. *Let $X_1, X_2, \ldots$ be random variables with $\mathbb{E}(X_n) > 0$ for every n. If $\frac{\mathrm{Var}(X_n)}{\mathbb{E}(X_n)^2} \to 0$, then $\Pr(X_n \neq 0) \to 1$.*

Proof. If $X_n = 0$ then $|X_n - \mathbb{E}(X_n)| = \mathbb{E}(X_n)$, so $\Pr(X_n = 0) \leq \Pr(|X_n - \mathbb{E}(X_n)| \geq \mathbb{E}(X_n))$, which is at most $\frac{\mathrm{Var}(X_n)}{\mathbb{E}(X_n)^2}$ by Theorem 1.3.3. □

Here is the smallest example worth doing, and the only one in which we will get a good bound out of Chebyshev alone.

Example 1.3.5. Flip n fair coins and let X be the number of heads, so that X is a sum of n independent random variables each taking the value 1 or 0 with probability $\frac{1}{2}$. Each of them has mean $\frac{1}{2}$ and variance $\frac{1}{4}$, so $\mathbb{E}(X) = \frac{n}{2}$ and, by additivity, $\mathrm{Var}(X) = \frac{n}{4}$. Theorem 1.3.3 at $a = \varepsilon n$ then gives

$$\Pr\left(\left|X - \frac{n}{2}\right| \geq \varepsilon n\right) \leq \frac{\mathrm{Var}(X)}{\varepsilon^2 n^2} = \frac{1}{4\varepsilon^2 n}$$

for every $\varepsilon > 0$, which tends to 0 as $n \to \infty$. So all but a vanishing proportion of the outcomes have between $\left(\frac{1}{2} - \varepsilon\right) n$ and $\left(\frac{1}{2} + \varepsilon\right) n$ heads.

1.4 Asymptotic Notation

Most of what we prove about random graphs holds only in the limit, and we need a compact way of saying how fast one quantity grows compared with another. Throughout this section, f and g are real functions of a natural number n, with $g(n) > 0$ for all large enough n, and every limit is as $n \to \infty$. The variable will not always be called n—$m(k) = O\left(k^2 \cdot 2^k\right)$ in Chapter 2 is a statement about $k \to \infty$—but there is always exactly one variable that tends to infinity, and everything else either is fixed or depends on it. In $G_{n,p}$, for instance, p is allowed to depend on n.

1.4.1 Little o and Little ω

The strongest comparison is that one function becomes negligible next to the other.

Definition 1.4.1 (Little o and Little ω). We write $f = o(g)$ if $\frac{f(n)}{g(n)} \to 0$, and $f = \omega(g)$ if $\frac{f(n)}{g(n)} \to \infty$.

In particular, $f = o(1)$ says that $f \to 0$, and $f = \omega(1)$ that $f \to \infty$. Unwinding the limit, $f = o(g)$ says that for every $\varepsilon > 0$ we eventually have $|f(n)| \leq \varepsilon\, g(n)$: however small a fraction of g we allow, f eventually fits inside it. When f and g are both eventually positive, $f = \omega(g)$ if and only if $g = o(f)$.

The equals sign here is a figure of speech. $o(g)$ is a class of functions, and $f = o(g)$ says that f belongs to it; $\frac{1}{n} = o(1)$ and $\frac{1}{n^2} = o(1)$ do not make $\frac{1}{n} = \frac{1}{n^2}$. For the same reason, $o(g)$ and $\omega(g)$ may appear inside a larger expression, where they stand for some function in the class. This is how Chapter 4 uses them.

Example 1.4.2. For the number X of triangles in $G_{n,p}$, $\mathbb{E}(X) = \binom{n}{3}p^3$, which is roughly $\frac{(np)^3}{6}$. So if $p = o\left(\frac{1}{n}\right)$ then $np \to 0$ and $\mathbb{E}(X) \to 0$, while if $p = \frac{\omega(1)}{n}$, ie, $np \to \infty$, then $\mathbb{E}(X) \to \infty$. Similarly, the hypothesis of Corollary 1.3.4 is that $\mathrm{Var}(X_n) = o\left(\mathbb{E}(X_n)^2\right)$.

The clique number $\omega(G)$ of Chapter 2 shares a letter with this notation. They are never confused in practice: the clique number takes a graph, and little ω takes a function.

1.4.2 Big O and Big Ω

Often all we can say, or all we need, is that one function is at most a constant multiple of another.

Definition 1.4.3 (Big O and Big Ω). We write $f = O(g)$ if there are $C > 0$ and n_0 such that $|f(n)| \leq Cg(n)$ for all $n \geq n_0$, and $f = \Omega(g)$ if there are $c > 0$ and n_0 such that $f(n) \geq c\,g(n)$ for all $n \geq n_0$.

The same conventions apply: these are classes of functions, the equals sign means membership, and when f and g are both eventually positive, $f = \Omega(g)$ if and only if $g = O(f)$. Each is weaker than its little counterpart: $f = o(g)$ implies $f = O(g)$, and likewise $f = \omega(g)$ implies $f = \Omega(g)$, but $2n = O(n)$ and $2n \neq o(n)$.

Example 1.4.4. Proposition 6.2.8 says that $\mathrm{VW}(k, 2) > \frac{(k-1)\,2^{k-1}}{ek^2}$ for all $k \geq 2$. Since $\frac{k-1}{k} \geq \frac{1}{2}$, the right-hand side is at least $\frac{1}{4e} \cdot \frac{2^k}{k}$, so $\mathrm{VW}(k, 2) = \Omega\left(\frac{2^k}{k}\right)$, with $c = \frac{1}{4e}$, for all $k \geq 2$. The exercise after Theorem 2.1.10, that $m(k) = O\left(k^2 \cdot 2^k\right)$, asks for a bound from above instead: some C with $m(k) \leq Ck^2 \cdot 2^k$ for all large k.

With this we end our terse review.

Chapter 2

Graphs and Colourings

Much of this course will be about graphs, and a surprising amount of what we want to know about a graph can be phrased as a question about colouring it. We begin with the vocabulary, and then settle the first such question completely: which graphs admit a colouring with only two colours. We then ask the same question of hypergraphs, where it becomes considerably harder—even pinning down how many edges a 3-uniform hypergraph needs before two colours are no longer enough takes real work. We then colour graphs with as many colours as it takes, and find that the number needed can be forced up as far as we like without ever creating a short cycle. Last, we colour the plane itself, an infinite graph whose chromatic number we can only trap between 4 and 7. The hypergraphs get a chapter of their own after that, and the opposite question—how large a complete graph must be before every colouring of its edges is forced to produce something monochromatic—gets one too, Chapter 6.

Notation.

- $[k]$ will denote $\{1, \dots, k\}$.
- $\sim$ will denote the adjacency relation on graph vertices.
- K_n will denote the complete graph on n vertices.
- $\mathcal{H}$ will denote the set of hyper-edges of a hypergraph H, and likewise for any other letter; calligraphic letters will denote families of sets more generally.

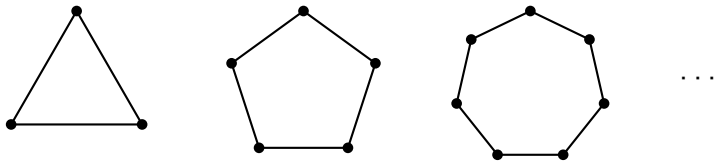
2.1 Colourings

Definition 2.1.1 (Colouring). Given a graph $G = (V, E)$, a k-colouring of G is a map $c : V \to [k]$ such that $u \sim v \implies c(u) \neq c(v)$.

2.1.1 2-Colourings of Graphs

Let's start with a motivating question: which graphs can be 2-coloured?

Any odd cycle graph can't be 2-coloured. That is, any graph that looks like



Let's study when odd cycles exist.

Lemma 2.1.2. *If G has an odd closed walk, then G has an odd cycle.*

Recall that a closed walk need not be a cycle: walks can contain repetitions. So even line graphs can have closed walks (start from one vertex, go to its neighbour, return).

Proof. Suppose G has an odd closed walk. Let W be such a walk of minimal length. Write $W = (x_1, \ldots, x_i, \ldots, x_j, \ldots, x_\ell)$, with $x_\ell = x_1$. Suppose W is not a cycle. Then there must be some x_i, x_j such that $x_i = x_j$ with $1 \leq i < j \leq \ell - 1$, ie, there must be some repetition beyond the closing one. Then, $(x_1, \ldots, x_i, x_{j+1}, \ldots, x_\ell)$ and $(x_i, x_{i+1}, \ldots, x_j)$ are both shorter walks than W. Moreover, one of them must be of odd length. This contradicts minimality of the length of W. So W must be a cycle. □

Here's a cool result.

Theorem 2.1.3. *A graph $G = (V, E)$ can be 2-coloured if and only if G has no odd cycle.*

Proof.

($\implies$) We've already seen that odd cycles can't be 2-coloured. So if G can be 2-coloured, it can't have an odd cycle. We've essentially seen this from the examples above.

($\impliedby$) We may assume G is connected, since otherwise we can colour each component separately. Fix some $v_0 \in V$. Define $c : V \to \{1, 2\}$ by

$$c(u) = \begin{cases} 2 & \text{if } \operatorname{dist}(v_0, u) \text{ is odd} \\ 1 & \text{if } \operatorname{dist}(v_0, u) \text{ is even} \end{cases}$$

where $\operatorname{dist}(\cdot, \cdot)$ denotes the shortest number of edges from one vertex to another.

We claim that this c is a valid 2-colouring.

Indeed, suppose it is not. Then, there must be some $u, v \in V$ such that $u \sim v$ and $c(u) = c(v)$. Now let's consider the two possible cases on the colour.

Suppose $c(u) = c(v) = 1$ for adjacent u, v. Then, there is a path of even length between v and v_0 and another path of even length between u and v_0. Concatenating the path from v to v_0 with the one from v_0 to u, and closing up with the edge uv, we get a closed walk of length even + even +1. This is odd, so by Lemma 2.1.2, G has an odd cycle.

The case $c(u) = c(v) = 2$ is identical: the two paths are now of odd length, so the closed walk they form with the edge uv has length odd + odd +1, which is odd again.

Either way, G has an odd cycle, contrary to hypothesis. So c must be a valid 2-colouring.

□

2.1.2 Colourings of Hypergraphs

Colouring is not really a question about graphs. All the definition above asks of G is that we know which pairs of vertices may not share a colour, and nothing stops us forbidding larger sets instead.

Definition 2.1.4 (Hypergraph). Given a vertex set V, a **hypergraph** is a pair $H = (V, \mathcal{H})$, where $\mathcal{H}$ is a collection of subsets of V. The sets in $\mathcal{H}$ are called **hyper-edges**.

Definition 2.1.5 (Uniformity). A hypergraph H is k-**uniform** if $|f| = k$ for all $f \in \mathcal{H}$.

The following is thus obvious.

Proposition 2.1.6. *A graph is a 2-uniform hypergraph.*

Definition 2.1.7. A proper colouring of a hypergraph H is an assignment $c : V \to [r]$ such that $\forall f \in \mathcal{H}$, $|c(f)| \geq 2$.

We say "no edge is monochromatic".

We can ask ourselves the following question: **"Which 3-uniform hypergraphs are 2-colourable?**

For each k, define $m(k)$ to be the smallest m such that $\exists$ some k-uniform hypergraph with m edges that is not 2-colourable. We can show that $m(2) = 3$. But what is $m(3)$? That is, how many 3-edges do I need to prevent 2-colourability?

2.1.3 The Value of $m(3)$

Let H be a 3-uniform hypergraph that's not 2-colourable. Assume, further, that H has at most 6 edges (each being a subset of size 3).

First, we'll show that we can reduce, without loss of generality, to the case where H has exactly 6 vertices. Indeed, if H has vertices x, y that are not in any common edge, then I can define H' by "merging" x and y. So H' has one fewer vertex than H. This H' cannot be 2-colourable: if it were, then that colouring would extend to H by using the colour of the merged vertex at both x and y, which would make H 2-colourable. And when H has more than 6 vertices, such a pair always exists.

An edge has 3 vertices, so it covers the $\binom{3}{2} = 3$ pairs among them, and H has at most 6 edges, so at most

$$6 \cdot \binom{3}{2} = 18$$

pairs are covered in total. But if H has $n \geq 7$ vertices, there are $\binom{n}{2} \geq \binom{7}{2} = 21$ pairs to cover.

So some pair is left uncovered, and merging it brings us one vertex closer to 6.

So now assume that H has exactly 6 edges and 6 vertices and is not 2-colourable.

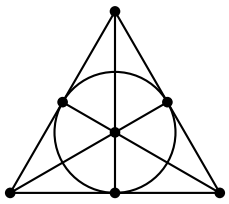
There is no loss in the vertex count in the other direction either: a hypergraph with fewer than 6 vertices can be handed as many extra vertices as it likes, and vertices lying in no edge change nothing about which colourings are proper.

Now count colourings. Among all the ways of colouring 6 vertices with two colours, consider only the balanced ones, which split V into two sets of three; there are $\frac{1}{2}\binom{6}{3} = 10$ of these. Such a colouring fails exactly when some edge is monochromatic, and an edge of H has three vertices, so it is monochromatic under a balanced colouring precisely when it is one of the two colour classes. Each edge f therefore rules out exactly one of the ten balanced colourings, namely the one splitting V into f and $V \setminus f$. Six edges rule out at most six of the ten, so at least four survive, and any one of them is a proper 2-colouring of H. This contradicts our assumption, so no such H exists and

$$m(3) \geq 7$$

That bound is attained, and the hypergraph attaining it is a familiar object.

Example 2.1.8 (The Fano Plane)**.** The **Fano plane** is the 3-uniform hypergraph on the seven vertices below, whose seven edges—by convention, its **lines**—are the three sides of the triangle, the three medians, and the circle through the three edge midpoints.



Every vertex lies on exactly three lines, any two vertices lie on exactly one line together, and any two lines meet in exactly one vertex. That last property is what does the work below.

Proposition 2.1.9. *The Fano plane is not* 2*-colourable. Hence* $m(3) = 7$*.*

Proof. Of the $\binom{7}{4} = 35$ sets of four vertices, exactly 28 contain a line: a line together with any one of the four vertices off it gives such a set, which is $7 \cdot 4 = 28$ sets in all, and no two of them coincide because four vertices cannot contain two lines—two lines meet in a vertex, so between them they cover five. The seven sets left over are the complements of the seven lines.

Now suppose we had a proper 2-colouring, with colour classes A and B, and say $|A| \leq 3$. If $|A| = 3$, then A is not a line, since A is monochromatic; so $B = V \setminus A$ is a set of four vertices that is not the complement of a line, and hence contains one. If $|A| \leq 2$, then the vertices outside B lie on at most $3 + 3 - 1 = 5$ lines between them, so at least two lines avoid them both and lie inside B. Either way some line is monochromatic, which is a contradiction.

So the Fano plane is a 3-uniform hypergraph with seven edges that is not 2-colourable, giving $m(3) \leq 7$. □

2.1.4 Bounds on $m(k)$

Pinning down $m(3)$ took a page of case analysis, and doing the same for $m(4)$ is out of the question. What we can do for every k at once is bound it, and the lower bound falls out of an idea that will keep coming back: colour at random, and show that a bad colouring is improbable.

Theorem 2.1.10. *$m(k) \geq 2^{k-1}$. In other words, any k-uniform hypergraph with $< 2^{k-1}$ edges is 2-colourable.*

Proof. Given a hypergraph with $< 2^{k-1}$ edges, randomly colour the vertices red and blue, with equal probability (1/2) for each. For each edge f, let $\mathcal{E}_f$ be the event that f is monochromatic. Then,

$$\Pr(\mathcal{E}_f) = \frac{2}{2^k}$$

Taking a simple union bound,

$$\Pr\left(\bigcup_{f \in \mathcal{H}} \mathcal{E}_f\right) \leq \frac{|\mathcal{H}|}{2^{k-1}} < 1$$

where we interpret $\bigcup_{f \in \mathcal{H}} \mathcal{E}_f$ as the event that there is a monochromatic edge. As the probability

is less than 1, there must be at least one colouring with no monochromatic edge. □

Note that the above argument can also be written combinatorially instead of probabilistically. In this instance, the probabilistic language essentially conveys the same information as combinatorial language would. But this may not always be true, so it's worth being careful about this.

In the other direction, the truth is only a factor of about k^2 away.

Exercise 2.1.11. Prove that $m(k) = O\left(k^2 \cdot 2^k\right)$

2.1.5 Maker-Breaker Games

Theorem 2.1.10 says that a hypergraph with few enough edges has a proper 2-colouring somewhere among all the colourings of its vertices. It turns out that the very same threshold survives a far more demanding requirement: that one of two players, claiming vertices one at a time, be able to stop the other from ever owning a whole edge. Winning that does not by itself produce a proper 2-colouring, since the winner's own class may contain a monochromatic edge. It does force one to exist all the same: a breaker who can win going second can also win going first, by holding a spare vertex in reserve, and setting those two strategies against each other leaves neither class containing an edge.

Here's a fun fact: a "maker-breaker" game is a game where the "maker" wins if they are able to achieve an objective and the "breaker" wins if they are able to stop the maker from winning. So if you set up a "maker-breaker" version of tic-tac-toe, where only the crosses can win by making three in a row and the noughts can only win by stopping the crosses, the crosses will always win (assuming the crosses play first). This is, of course, different from tic-tac-toe, which, as our own studies from when we were a kid have shown, can always be drawn.

In the language of hypergraphs, the game is played on the vertices of H: the two players alternately claim unclaimed vertices, maker wins by claiming every vertex of some edge, and breaker wins if play ends with no edge wholly maker's.

Whether breaker can hold out depends not only on how many edges there are, but on how many of them pass through a single vertex.

Lemma 2.1.12. *Let H be a k-uniform hypergraph with m edges and maximum degree*

$$\Delta = \max_{v \in V} |\{f \in \mathcal{H} \mid v \in f\}|$$

If $m + \Delta < 2^k$, then breaker wins (if breaker goes second).

Proof. At the start of move i, we will conceptualise the danger of edge $f \in \mathcal{H}$ by

$$\phi_i(f) = \begin{cases} 2^{\text{number of maker moves in } f} & \text{if there are no breaker moves in } f \\ 0 & \text{if there is a breaker move in } f \end{cases}$$

$$\Phi_i = \sum_{f \in \mathcal{H}} \phi_i(f)$$

Consider the strategy where, once maker has played y on move i, breaker plays the unclaimed vertex carrying the most danger, that is, the x maximising

$$\sum_{f \ni x} \phi(f)$$

with ϕ measured after maker's move. Two things move Φ on move i. Maker's play doubles the danger of every edge through y that breaker has not already killed, so it raises Φ by $a_i = \sum_{f \ni y} \phi_i(f)$; breaker's play sends the danger of every surviving edge through x to 0, so it lowers Φ by the quantity b_i that breaker has just maximised. Hence

$$\Phi_{i+1} = \Phi_i + a_i - b_i$$

The point of breaker's rule is that $a_i \leq b_{i-1}$ for every $i \geq 2$. Indeed, y is unclaimed right up until maker takes it on move i, so it was one of the vertices breaker was choosing between on move $i-1$, and breaker chose one carrying at least as much danger as it. Between then and the start of move i nothing happens except breaker's own play, which only ever sends dangers to 0, so the danger through y at the start of move i is no greater than it was when breaker passed y over.

Summing $\Phi_{i+1} = \Phi_i + a_i - b_i$ over the first n moves and cancelling each a_i against b_{i-1}, everything past the first two terms disappears and we are left with

$$\Phi_{n+1} \leq \Phi_1 + a_1$$

Stopping the same sum one term earlier gives the same bound halfway through a move, after maker has played and before breaker has replied. Now $\Phi_1 = m$, since at the start every edge contains a move of neither player and so contributes $2^0 = 1$, and a_1 is the number of edges through maker's opening vertex, which is at most Δ. So the danger never exceeds $m + \Delta$ at any point in the game.

But if maker ever completes an edge f, then at that moment f contains all k of its vertices from maker and none from breaker, so $\phi(f) = 2^k$ and hence $\Phi \geq 2^k$. Since $m + \Delta < 2^k$, this never happens, and breaker wins. □

No vertex lies in more edges than there are, so $\Delta \leq m$, and the threshold at which breaker can hold out is exactly the one we have just been looking at.

Corollary 2.1.13 (Erdős-Selfridge). *If H is a k-uniform hypergraph with m edges, and $m < 2^{k-1}$, then breaker wins (if breaker goes second).*

Proof. $m + \Delta \leq 2m < 2^k$, so Lemma 2.1.12 applies. □

Let's consider higher-dimensional maker-breaker tic-tac-toe.

Example 2.1.14. Turns out that $8 \times 8 \times 8$ maker-breaker tic-tac-toe is breaker-winnable: we can show that $m = \frac{10^3 - 8^3}{2} = 244$ and $\Delta = 7$, meaning $m + \Delta = 251 < 2^k = 256$.

Both numbers come out of the geometry. For m, pad the board with one extra cell in every direction, giving a $10 \times 10 \times 10$ cube, and run each winning line one step past each of its two ends. The two cells we land on lie in the shell we just added, and every cell of the shell is the end of exactly one line: a coordinate of the cell equal to 0 tells the line to increase along that axis, one equal to 9 tells it to decrease, and the rest stay fixed. So the lines are in bijection with those pairs of shell cells that arise as the two ends of a line, and since every shell cell is the end of exactly one line there are $\frac{10^3 - 8^3}{2}$ of them. For Δ, a line through a given cell is determined by its direction, and a corner cell has the 3 lines along the axes, the 3 face diagonals and the one long diagonal through it, which is 7; no cell of an $8 \times 8 \times 8$ board does better. And it is Lemma 2.1.12 doing the work here rather than the corollary, since 244 is a long way above $2^{k-1} = 128$.

We won't say any more about tic-tac-toe in this course, but it was a fun aside.

2.2 Chromatic Number

Back to colouring vertices, and this time we are not rationed to two colours. Chapter 2 opened by asking which graphs can be coloured with two of them; the question now is how many colours a graph needs in general, and how far that number can be pushed away from the obvious lower bound.

Definition 2.2.1 (Chromatic Number)**.** The **chromatic number** of a graph G, denoted $\chi(G)$, is the smallest number of colours I need to colour G.

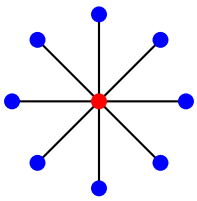
2.2.1 Trivial Bounds

There are some trivial bounds on $\chi(G)$.

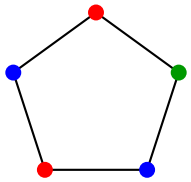
Definition 2.2.2 (Clique)**.** A **clique** in G is a complete subgraph, ie, a set of vertices any two of which are adjacent. The **clique number** of a graph, denoted $\omega(G)$, is the number of vertices in the largest clique.

It isn't hard to see that $\omega(G) \leq \chi(G)$ always. Similarly, it is clear that $\chi(G) \leq \Delta + 1$, where Δ is the maximum degree.

The upper-bound is absolutely not tight, for instance we can have a "star graph" with one vertex in the middle connected to a number of other vertices. This has high Δ but $\chi(G) = 2$.



Similarly, the lower-bound is also not tight. For example, any odd cycle on more than 3 vertices (eg, the 5-cycle) has clique number 2 but chromatic number 3.



In fact, we will show that $\chi(G)$ can be much bigger than $\omega(G)$.

2.2.2 Perfect Graphs

First, the graphs for which the lower bound is exact—and stays exact when we throw vertices away. That needs a word for what is left when we do.

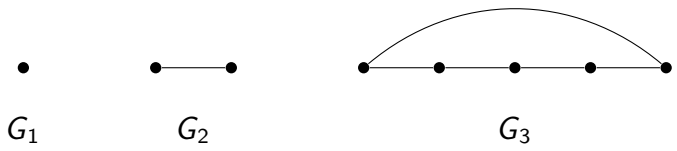
Definition 2.2.3 (Induced Subgraph). Given a graph $G = (V, E)$ and a subset $W \subseteq V$, the subgraph of G **induced** by W, denoted $G[W]$, is the graph with vertex set W whose edges are precisely the edges of G with both endpoints in W. An **induced subgraph** of G is any subgraph of the form $G[W]$.

Definition 2.2.4 (Perfect Graph). A graph G is **perfect** if $\chi(H) = \omega(H)$ for all induced subgraphs H of G.

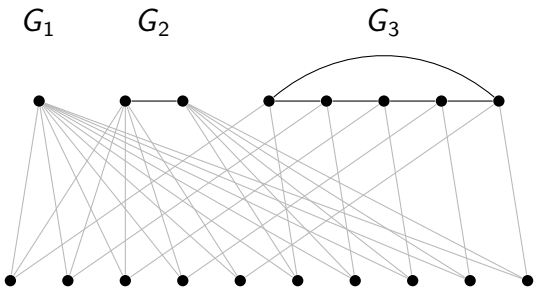
2.2.3 Zykov's Construction

At the opposite extreme are graphs with $\omega = 2$ and χ as large as we like, and we can build them by hand.

Let's now construct a sequence $(G_n)_{n\in\mathbb{N}}$. Start with G_1 a single vertex. Given $G_1, \ldots, G_{k-1}$, take disjoint copies of all of them, and for every way of choosing one vertex v_i from each G_i add a new vertex adjacent to $v_1, \ldots, v_{k-1}$ and to nothing else; the result is G_k. There are $|G_1| \cdots |G_{k-1}|$ new vertices, so G_2 has $1+1=2$ vertices and is a single edge, G_3 has $1+2+2=5$ and is the 5-cycle, and G_4 has $1+2+5+10=18$ (the construction is due to Zykov).



G_4 is already hard to take in, but the shape of the construction survives in it: the three earlier graphs across the top, and below them one new vertex for each of the $1 \cdot 2 \cdot 5 = 10$ ways of picking one vertex from each. The edges into the copies are drawn faintly, so that the copies themselves stay visible.



These have exactly the property we want.

Proposition 2.2.5. *For every $k \geq 1$, G_k has no triangles and $\chi(G_k) = k$.*

Proof. Both by induction on k, and both trivial for G_1. The new vertices of G_k form an independent set, and the neighbours of any one of them lie one in each of $G_1, \ldots, G_{k-1}$, so no two of them are adjacent. A triangle can therefore contain at most one new vertex, and if it contains one then its other two vertices are adjacent neighbours of that vertex, which do not exist. So every triangle of G_k lies inside some G_i, and by induction there are none.

For $\chi(G_k) \leq k$, colour each G_i with $i \leq k-1$ colours, which we can by induction, and give every new vertex colour k. For $\chi(G_k) \geq k$, suppose c is a proper $(k-1)$-colouring of G_k. Its restriction to G_i is a proper colouring of G_i, and so uses at least $\chi(G_i) = i$ colours. Pick $v_1 \in G_1$; G_2 uses at least two colours, so pick $v_2 \in G_2$ with $c(v_2) \neq c(v_1)$; G_3 uses at least three, so pick $v_3 \in G_3$ with $c(v_3) \notin \{c(v_1), c(v_2)\}$; and so on, at each step choosing $v_i \in G_i$ to avoid the $i-1$ colours chosen so far. After $k-1$ steps, $c(v_1), \ldots, c(v_{k-1})$ are $k-1$ distinct colours, which is all of them, and the new vertex attached to $v_1, \ldots, v_{k-1}$ has no colour left. □

2.2.4 Graphs of Large Girth

The graphs of Proposition 2.2.5 have no triangles, but nothing stops them having cycles of length four or five—G_3 is one. It is natural to ask whether forbidding all short cycles finally holds the

chromatic number down, and the answer is that it does not.

Definition 2.2.6 (Girth). The **girth** of G, denoted girth(G), is the length of the shortest cycle in G.

The girth of a graph with no cycles is infinite.

Theorem 2.2.7 (Erdős). *For any k, there are graphs of girth at least k and chromatic number at least k.*

The proof runs through independent sets rather than through colourings directly.

Definition 2.2.8 (Independence Number). A set S of vertices of G is **independent** if no two of its elements are adjacent. The **independence number** of G, denoted $\alpha(G)$, is the size of the largest independent set in G.

We know we can construct graphs of average degree d with $\frac{nd}{2}$ edges. We want to prove that such a graph has a "large" independent set.

As a first attempt, let's choose a set S randomly by including each vertex of G independently with probability p. Write $e(S)$ for the number of edges of G with both endpoints in S, so that S is independent exactly when $e(S) = 0$. Then,

$$\mathbb{E}(|S|) = pn$$
$$\mathbb{E}(e(S)) = \frac{p^2 nd}{2}$$

the second line because each of the $\frac{nd}{2}$ edges lands inside S with probability p^2. We want S large and $e(S)$ small, and the two pull against each other: the expected size grows like p but the expected number of edges inside like p^2. So we take p as large as we can while keeping the expected number of edges inside S down to about one. I would want $p \simeq \sqrt{\frac{2}{dn}}$. In that case we would get something like $\mathbb{E}(|S|) \approx \sqrt{\frac{n}{d}}$. This is a poor return: a disjoint union of copies of K_{d+1} has average degree d and an independent set of size $\frac{n}{d+1}$, and the trouble is that we have insisted on S coming out independent all by itself.

Now, let's make another attempt. Still choose S randomly, including each vertex with probability

p.

Let $X = |S|$ and $Y = e(S)$.

Now consider $\mathbb{E}(X - Y)$. This is $pn - \frac{p^2 nd}{2}$ by linearity of expectation. So if S has X vertices and Y edges inside it, then G contains an independent set of size at least $X - Y$, and using $p = \frac{1}{d}$ gives $\mathbb{E}(X - Y) = \frac{n}{2d}$.

This needs $d \geq 1$ for $p = \frac{1}{d}$ to be a probability at all, and then some S has $X - Y \geq \mathbb{E}(X - Y) = \frac{n}{2d}$. Now alter that S: for each of the Y edges lying inside it, throw out one of the two endpoints. At most Y vertices go, and no edge of G keeps both its endpoints, so what is left is an independent set of size at least $X - Y \geq \frac{n}{2d}$.

What we have shown is worth recording.

Lemma 2.2.9. *If G is a graph with n vertices and average degree $d \geq 1$, then G has an independent set of size at least $\frac{n}{2d}$.*

The above is called the **method of alterations**—constructing something randomly and then changing based on how it goes.

The proof of Theorem 2.2.7 plays the same trick, but with a random graph in place of a random set of vertices.

Definition 2.2.10 (Random Graph)**.** For $n \in \mathbb{N}$ and $p \in [0, 1]$, the **random graph** $G_{n,p}$ is the graph on the vertex set $[n]$ in which each of the $\binom{n}{2}$ possible edges is present independently with probability p.

Concretely, the probability space we are in is the set Ω of all graphs on the vertex set $[n]$, of which there are $2^{\binom{n}{2}}$. An event is any collection of such graphs—the graphs containing a triangle, say, or those with an independent set of size a—and the probability measure is given by the value of p: a graph G on $[n]$ with $e(G)$ edges has probability

$$\Pr(G_{n,p} = G) = p^{e(G)} (1 - p)^{\binom{n}{2} - e(G)}$$

This is the finite setting of Section 1.1, with every collection of graphs an event. In particular, when we study the probability of some event, we are effectively counting the graphs in which the desired outcome is realised, each weighted as above. When $p = \frac{1}{2}$ every graph has the same weight $2^{-\binom{n}{2}}$, and the probability of an event is exactly the proportion of all graphs on $[n]$ that lie in it.

Proof of Theorem 2.2.7. Fix $k \geq 3$; for $k \leq 2$ a single edge already does. Consider the graph $G_{n,p}$ for $p = \frac{n^\epsilon}{n}$. For some small $\epsilon > 0$ (in particular, $\epsilon \ll \frac{1}{k}$), we see that

$$\begin{aligned}\Pr(\alpha(G_{n,p}) \geq a) &\leq \binom{n}{a}(1-p)^{\binom{a}{2}} \\ &\leq n^a e^{-p\binom{a}{2}} = \exp\left(a\left(\ln n - \frac{p\,(a-1)}{2}\right)\right)\end{aligned}$$

using $\binom{n}{a} \leq n^a$ and $1-p \leq e^{-p}$. Take $a = \left\lceil \frac{n}{2k} \right\rceil$, so that $a-1 \geq \frac{n}{2k} - 1$ and hence $\frac{p(a-1)}{2} \geq \frac{n^\epsilon}{4k} - \frac{1}{2}$. Any positive power of n eventually beats $\ln n$, so the bracket is negative for n large enough and the whole thing tends to 0: that is, $\Pr\left(\alpha(G_{n,p}) \geq \frac{n}{2k}\right) \to 0$ as $n \to \infty$.

Now, let X be the number of short cycles. That is, cycles of length $\leq k$. Then,

$$\begin{aligned}\mathbb{E}(X) &= \sum_{\ell=3}^{k} \frac{\binom{n}{\ell} \cdot \ell!}{2\ell} p^\ell \\ &\leq \frac{1}{2}\sum_{\ell=3}^{k} n^\ell p^\ell \\ &\leq (k-2)\,\frac{1}{2} n^k p^k \\ &\leq \frac{k}{2} n^{\epsilon k}\end{aligned}$$

By Markov's inequality (Theorem 1.3.1),

$$\Pr\left(X \geq \frac{n}{2}\right) \leq \frac{\mathbb{E}(X)}{n/2} \to 0$$

as $n \to \infty$. So, there is some graph G with n vertices, for sufficiently large n, such that $X \leq \frac{n}{2}$ and $\alpha(G) \leq \frac{n}{2k}$. Construct a new graph G' by deleting one vertex from each short cycle. This new graph has at least $\frac{n}{2}$ vertices, independence number at most $\frac{n}{2k}$, and thus chromatic number at least $\frac{n/2}{n/2k} \geq k$. □

2.3 The Chromatic Number of the Plane

Every graph so far has been finite. The plane gives us an infinite one to colour: we will trap its chromatic number between 4 and 7, and then see that only its finite subgraphs matter.

2.3.1 The Unit Distance Graph

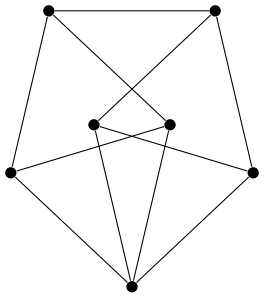
The graph in question has a vertex for every point of the plane.

Definition 2.3.1 (Unit Distance Graph). The **unit distance graph** on $V = \mathbb{R}^2$ is given by the adjacency relation $u \sim v$ iff $\|u - v\| = 1$.

Write G for this graph. What is $\chi(G)$?

One clear lower-bound on $\chi(G)$ is 3: G contains the equilateral triangle with side length 1 as a subgraph.

It turns out that there is a subgraph of G with chromatic number 4, known as Moser's Spindle.



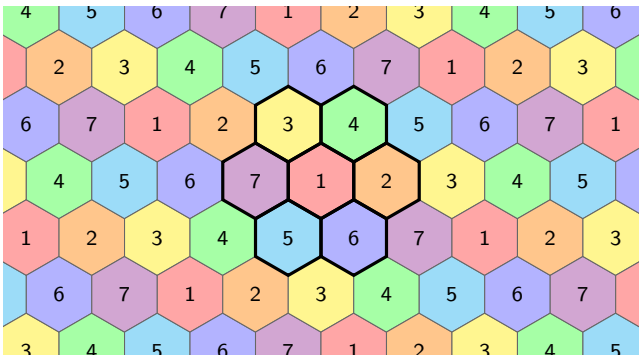
We thus conclude that $\chi(G) \geq 4$.

We can actually show that $\chi(G) \leq 7$. Indeed, consider the hexagonal tessellation of the plane. The point is that for a tessellation with hexagons of some diameter,

1. all points in any given hexagon are of the same colour
2. each hexagon and its 6 neighbours have different colours

Then, we can repeat colours *after* a unit of 7 (consisting of a hexagon and its neighbours).

Take the hexagons to have diameter d with $\frac{2}{\sqrt{21}-2} < d < 1$—say $d = 0.9$—and give each boundary point to just one of the hexagons it lies on. Two points of one hexagon are less than 1 apart, so giving a hexagon a single colour breaks nothing. The centres of neighbouring hexagons are $\frac{\sqrt{3}}{2}d$ apart, and in the colouring below the nearest hexagon of the same colour as a given one is two steps away in one direction and one in the next, so two same-coloured centres are at distance at least $\sqrt{7} \cdot \frac{\sqrt{3}}{2}d = \frac{\sqrt{21}}{2}d$. Every point of a hexagon is within $\frac{d}{2}$ of its centre, so two points of the same colour in different hexagons are at distance at least $\frac{\sqrt{21}}{2}d - d > 1$. No two points at distance exactly 1 share a colour, and $\chi(G) \leq 7$.



2.3.2 Compactness

Now for a topological detour!

Recall that a space is compact iff the closed sets satisfy the finite intersection property: for closed sets Y_α, with $\alpha \in I$, if $\bigcap_{\alpha\in J} Y_\alpha$ is nonempty for all finite $J \subseteq I$, then $\bigcap_{\alpha\in I} Y_\alpha$ is nonempty as well.

We'll use this property (and Tychonoff) to prove a 'compactness theorem' for the chromatic number of the plane.

Theorem 2.3.2. *For any graph G,*

$$\chi(G) = \sup\{\chi(H) \mid H \subseteq G \text{ and } H \text{ is finite}\}$$

First, a lemma.

Lemma 2.3.3. *If every finite subgraph B of G is k-colourable, then so is G.*

Proof. Consider $[k]$ with the discrete topology and $[k]^{V(G)}$ with the product topology. Note that $[k]^{V(G)}$ is the set of all (not necessarily proper) colourings of G.

For all $e \in E(G)$, define

$$P_e = \left\{ c \in [k]^{V(G)} \;\middle|\; c(u) \neq c(v) \text{ for } e = \{u, v\} \right\}$$

It is possible to show that P_e is clopen in $[k]^{V(G)}$: it is the preimage of the clopen set $\{(a, b) \in [k]^2 \mid a \neq b\}$ under the projection onto the coordinates u and v, which is continuous.

If every finite subgraph can be k-coloured, then for all finite $J \subseteq E(G)$, $\bigcap_{e \in J} P_e \neq \emptyset$: the edges in J span a finite subgraph, a proper k-colouring of it extends to all of $V(G)$ by colouring the remaining vertices arbitrarily, and the result lies in every P_e with $e \in J$. By Tychonoff's theorem $[k]^{V(G)}$ is compact, the P_e are closed, and so the finite intersection property gives $\bigcap_{e \in E(G)} P_e \neq \emptyset$. Any c in that intersection has $c(u) \neq c(v)$ for every edge $\{u, v\}$, ie, is a proper k-colouring of G. □

Proof of Theorem 2.3.2. Every subgraph H of G has $\chi(H) \leq \chi(G)$, so the supremum is at most $\chi(G)$. Conversely, if the supremum is some finite k, then every finite subgraph is k-colourable, so G is by Lemma 2.3.3, and $\chi(G) \leq k$. □

Chapter 3

Set Systems

Strip the colouring question away from a hypergraph and what is left is a family of subsets of a finite set, and this chapter asks what such a family can look like. Three questions, each with a theorem for an answer: whether one vertex can be chosen out of each edge, all of them different, which is Hall's marriage theorem; how many edges there can be once no edge is allowed to sit inside another, which is Sperner's; and how many there can be once no two edges are allowed to miss each other altogether, which is Erdős-Ko-Rado. The last two are answered by the same device, a permutation chosen uniformly at random, and it is worth watching for it the second time round.

3.1 Matchings in Bipartite Graphs

Hypergraphs came up in Definition 2.1.4 as the setting in which the colouring question gets hard, and Chapter 2 set them aside again once it had its answer. The first thing this chapter asks of them is this: given a hypergraph, can we pick one vertex out of each edge, all of them different? Getting to the answer means first noticing that a hypergraph, a $0,1$ matrix and a bipartite graph are three views of the same data.

3.1.1 Adjacency Matrices

We can represent the data of a hypergraph H in a matrix, known as the adjacency matrix.

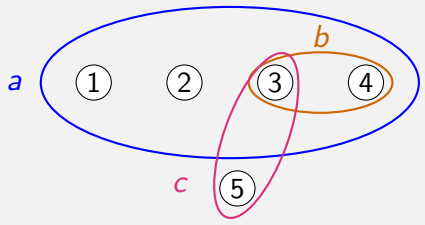
Definition 3.1.1 (Adjacency Matrix). If H is a hypergraph with n vertices and m hyper-edges, both taken in some fixed order, then the adjacency matrix of H, denoted Adj(H), is the unique $n \times m$ matrix (a_{ij}) where

$$a_{ij} = \begin{cases} 1 & \text{if vertex } i \text{ lies in hyper-edge } j \\ 0 & \text{otherwise} \end{cases}$$

Here are some examples of constructing adjacency matrices.

Example 3.1.2 (Some Adjacency Matrices of Hypergraphs).

1. Consider the hypergraph



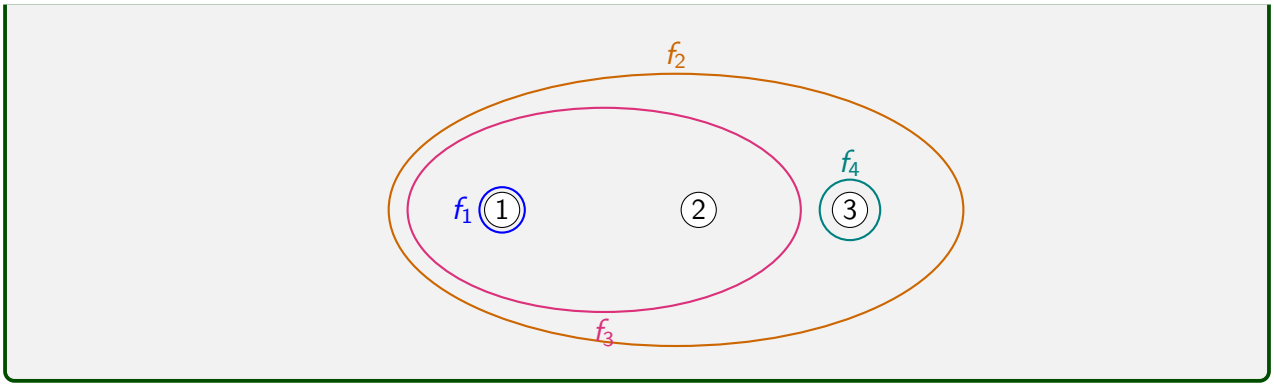
   Its adjacency matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

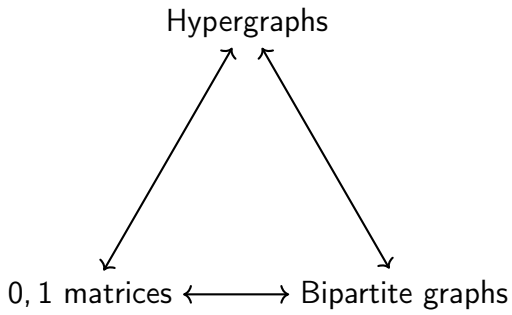
2. Consider the hypergraph

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

   Reading the columns off one at a time, this is the hypergraph on three vertices with the four hyper-edges $f_1 = \{1\}$, $f_2 = \{1, 2, 3\}$, $f_3 = \{1, 2\}$ and $f_4 = \{3\}$.



It turns out that there is not just a correspondence between hypergraphs and $0,1$ adjacency matrices, we can also complete the trifecta with bipartite graphs.



The bipartite graph attached to H has the vertices of H on one side and the hyper-edges of H on the other, with v joined to f exactly when $v \in f$, which is to say exactly when the adjacency matrix has a 1 in position (v, f).

3.1.2 Hall's Marriage Theorem

Recall, now, the most important result about bipartite graphs.

Theorem 3.1.3 (Hall's Marriage Theorem). *Given a bipartite graph with parts A and B, with A finite, there is a complete matching of A iff $\forall S \subseteq A$,*

$$|\Gamma(S)| \geq |S|$$

where $\Gamma(S) = \{v \in B \mid \exists u \in S \text{ s.t. } u \sim v\}$.

Proof.

($\implies$) Let M be a complete matching of A and let $S \subseteq A$. The partners under M of the elements of S are $|S|$ distinct vertices of B, each adjacent to something in S, so $\Gamma(S)$ has at least $|S|$ elements.

($\impliedby$) We induct on $|A|$. If $|A| = 1$, then Hall's condition applied to A itself says that the one vertex of A has a neighbour, and matching it to one is a complete matching. So suppose $|A| > 1$, and let's do some casework.

- Case 1: Assume that in fact, $\forall S \subsetneq A$ nonempty, $|\Gamma(S)| \geq |S| + 1$. Arbitrarily match some pair: pick any $a \in A$ and any $b \in \Gamma(\{a\})$, which is nonempty by hypothesis. After deleting this edge—and with it the vertices a and b and every other edge at either of them—the rest satisfies the desired result, since a nonempty $T \subseteq A \setminus \{a\}$ is a nonempty proper subset of A and so had at least $|T| + 1$ neighbours to begin with, of which deleting b can take away only one. What is left has $|A| - 1$ vertices on the left, so we are done by induction: it has a complete matching of $A \setminus \{a\}$, and adding the edge ab to that matching completes it to one of A.

- Case 2: Now assume the opposite, ie, that $\exists S \subsetneq A$ nonempty such that $|\Gamma(S)| = |S|$. We will apply induction again, on $|A|$ as before.

  If we restrict our attention to S, obviously we have a perfect matching between S and $\Gamma(S)$: every $T \subseteq S$ has $\Gamma(T) \subseteq \Gamma(S)$, so the graph they span satisfies Hall's condition and induction gives a complete matching of S, which the fact that they're of equal cardinality makes perfect. (There may be more than one, but there is one.)

  Define $A' = A \setminus S$, $B' = B \setminus \Gamma(S)$. We claim that the bipartite graph induced on A' and B' also has a complete matching of A', in which case we'd be done (by induction).

  Writing Γ' for the neighbourhood taken in that smaller graph, we have $\Gamma'(T) = \Gamma(T) \setminus \Gamma(S)$ for every $T \subseteq A'$, and $\Gamma(S \cup T) = \Gamma(S) \cup \Gamma(T)$ as it does for any two sets, so

  $$|\Gamma'(T)| = |\Gamma(S) \cup \Gamma(T)| - |\Gamma(S)| = |\Gamma(S \cup T)| - |S| \geq |S \cup T| - |S| = |T|$$

  where the inequality is Hall's condition applied to $S \cup T$, and the last step is that $T \subseteq A'$ is disjoint from S. (If $\Gamma(T)$ is infinite the display needs no reading at all: $\Gamma(S)$

is finite, so $\Gamma'(T)$ is infinite and certainly at least $|T|$.) So the smaller graph satisfies Hall's condition, and $|A'| < |A|$ because S is nonempty, so induction gives a complete matching of A' into B'. That matching touches no vertex of S and no vertex of $\Gamma(S)$, so laying it alongside the matching of S onto $\Gamma(S)$ gives a complete matching of A.

□

Given a hypergraph H, a family of distinct representatives is a collection of distinct $x_1, \ldots, x_m$ such that $x_i \in f_i$, where we have distinct hyper-edges $f_1, \ldots, f_m$ in $\mathcal{H}$. Such a collection exists iff for any subcollection $\mathcal{C}$ of $f_1, \ldots, f_m$,

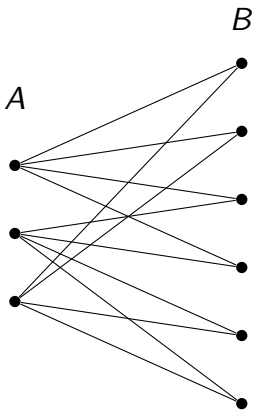
$$\left| \bigcup_{f \in \mathcal{C}} f \right| \geq |\mathcal{C}|$$

In other words, the number of vertices in all the edges in the collection should be at least the number of edges in the collection. This is what Hall's Marriage Theorem tells us about hypergraphs.

Hall's condition asks something of every subset of A, which is a lot to check. There is a consequence of it that only asks about degrees.

Theorem 3.1.4. *If G is a bipartite graph with parts A and B, if all degrees in A are d_1 and all degrees in B are d_2, with $1 \leq d_2 \leq d_1$, then there is a complete matching of A.*

Such a graph is a good deal more rigid than a bipartite graph with Hall's condition alone; here is one with $d_1 = 4$ and $d_2 = 2$.



Proof. Given a set $S \subseteq A$, there are $d_1 |S|$ edges coming out of S and going to the right. Each

vertex of $\Gamma(S)$ absorbs at most d_2 of them, so

$$|\Gamma(S)| \geq \frac{d_1 |S|}{d_2} \geq |S|$$

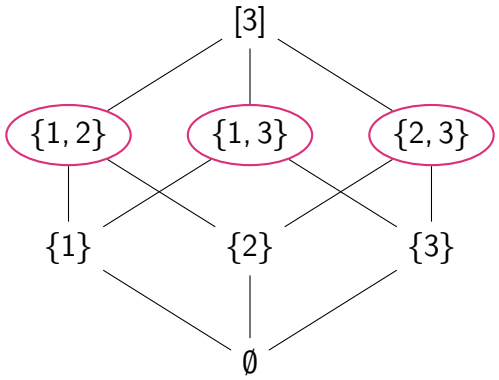
so we can just apply Theorem 3.1.3. □

3.2 Sperner Systems

A hypergraph is a vertex set together with a family of subsets of it, and the next question this chapter asks about one is how large that family can be. Unrestricted the question is dull—all 2^n subsets of $[n]$ form a hypergraph—so we forbid the cheapest way of making a family large, which is to pile sets up inside one another.

Definition 3.2.1 (Sperner System)**.** A collection of subsets of $[n]$ such that none is a subset of another is called a Sperner system.

Example 3.2.2. The three two-element subsets of [3], circled below, form a Sperner system.



The picture is the whole collection of subsets of [3] ordered by inclusion, arranged in layers by size, with an edge drawn from each set to every set one size up that contains it. Containment between two subsets is then visible as an upward path, so a Sperner system is precisely a collection of vertices no two of which are joined by such a path—and drawing the diagram turns a question about set inclusions into a question about which vertices we are allowed to pick. The diagram as a whole is emphatically not a Sperner system; each individual layer is, since two distinct sets of the same size never contain one another. That gives us two Sperner systems of size 3 here,

the singletons and the two-element subsets, and the top and bottom layers give two more, though useless ones: any system containing $\emptyset$ or $[3]$ can contain nothing else.

How large can a Sperner system be?

Theorem 3.2.3 (Sperner's Theorem). *A Sperner system on $[n]$ can have at most $\binom{n}{\lfloor n/2 \rfloor}$ sets.*

An intuitive way of arguing is to find complete matchings of the kth layer into the $(k+1)$th layer (cf. Theorem 3.1.4), but at some point, this starts to break down. And at what point is that? Before k exceeds $n/2$.

Instead, we go through a stronger inequality, weighting each set by how large its layer is.

Lemma 3.2.4 (LYM Inequality). *If $\mathcal{H}$ is a Sperner system on $[n]$, then*

$$\sum_{h \in \mathcal{H}} \frac{1}{\binom{n}{|h|}} \leq 1$$

Proof. Let $\mathcal{H}$ be a Sperner system on $[n]$. Define

$$\mathcal{F} = \{\emptyset, \{1\}, \{1,2\}, \{1,2,3\}, \ldots, \{1,2,\ldots,n\}\}$$

Choose a random permutation $\pi \in S_n$, and define

$$\mathcal{H}_\pi = \{\pi(h) \mid h \in \mathcal{H}\}$$

Define the random variable

$$X = |\{f \in \mathcal{F} \mid f \in \mathcal{H}_\pi\}|$$

Note that we can express

$$X = \sum_{h \in \mathcal{H}} \xi_{h,\pi}$$

where $\xi_{h,\pi}$ is 1 if $\pi(h) \in \mathcal{F}$ and 0 otherwise.

Computing the expected value of X, we get

$$\mathbb{E}(X) = \sum_{h \in \mathcal{H}} \mathbb{E}(\xi_{h,\pi})$$

Let's now compute the expectation of $\xi_{h,\pi}$. This is the probability that the permutation $\pi \in S_n$ sends $h \in \mathcal{H}$ to some element of $\mathcal{F}$. Since you cannot modify sizes, the only possible element in $\mathcal{F}$ to which π can send h is precisely the set $\{1, 2, \ldots, |h|\}$. How many such π can do this? We have to consider the way π acts on h, and the way it acts on the other $n - |h|$ elements of $[n]$, because we know it sends h to $\{1, 2, \ldots, |h|\}$. There are $|h|!\,(n - |h|)!$ such π, so the probability is

$$\frac{|h|!\,(n - |h|)!}{n!} = \frac{1}{\binom{n}{|h|}}$$

Now bound $\mathbb{E}(X)$ on the other side. Any π is a bijection and so preserves containment, which makes $\mathcal{H}_\pi$ a Sperner system; and $\mathcal{F}$ is a chain, any two of its members being comparable. A chain and a Sperner system share at most one set, so $X \leq 1$ for every π, and hence

$$1 \geq \mathbb{E}(X) = \sum_{h \in \mathcal{H}} \frac{1}{\binom{n}{|h|}}$$

□

Sperner's theorem is now a matter of replacing every term by the smallest it can be.

Proof of Theorem 3.2.3. Let $\mathcal{H}$ be a Sperner system on $[n]$. Since $\binom{n}{k}$ is largest at $k = \lfloor n/2 \rfloor$, Lemma 3.2.4 gives

$$1 \geq \sum_{h \in \mathcal{H}} \frac{1}{\binom{n}{|h|}} \geq \sum_{h \in \mathcal{H}} \frac{1}{\binom{n}{\lfloor n/2 \rfloor}} = \frac{|\mathcal{H}|}{\binom{n}{\lfloor n/2 \rfloor}}$$

and rearranging is exactly the bound we wanted. □

3.3 Intersecting Families

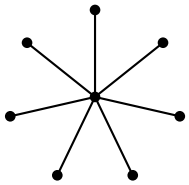
Sperner's theorem came out of forbidding one hyper-edge to sit inside another. Forbidding the opposite—a pair of hyper-edges that miss each other entirely—turns out to have a much more

interesting answer, and it is the one we spend this section on.

Definition 3.3.1 (Intersecting). We say that a k-uniform hypergraph H is **intersecting** if $\forall h_1, h_2 \in \mathcal{H},\ h_1 \cap h_2 \neq \emptyset$.

The question is: how big can an intersecting k-uniform hypergraph on n vertices be? Ie, how many hyper-edges can it have? Small k is worth doing by hand, and as long as $n \geq 2k$ every case points the same way—until the last one, where dropping that assumption is exactly the point.

- $k = 1$: we can see that the maximum number of edges must be 1.
- $k = 2$: the maximum number of edges is $n - 1$, achieved by the 'star graph'



- $k = 3$: $\binom{n-1}{2}$, achieved by fixing one vertex that lies in everything and choosing 2 more.
- $k \leq \frac{n}{2}$: $\binom{n-1}{k-1}$ by exactly the same strategy.
- $k > \frac{n}{2}$: we can do better! $\binom{n}{k}$ works because of something involving pigeons and holes.

3.3.1 The Erdős-Ko-Rado Theorem

The middle of that list is the general answer, and the bottom of it is why $n \geq 2k$ has to be assumed to get one.

Theorem 3.3.2 (Erdős-Ko-Rado). *If $n \geq 2k$, and H is an intersecting k-uniform hypergraph on n vertices, then the number of hyper-edges of H is $\leq \binom{n-1}{k-1}$. Moreover, if $n \geq 2k+1$, then the ("unique") extremal example is given by all k-subsets containing a fixed vertex.*

As a warmup case, let's assume that $k \mid n$. Suppose H is intersecting. Let $\mathcal{F}$ denote some partition

of $[n]$ into n/k subsets of size k (so $\mathcal{F}$ is fundamentally a collection of subsets).

$$X = |\{h \in \mathcal{H} \mid h = \pi(f) \text{ for some } f \in \mathcal{F}\}|$$

where $\pi \in S_n$ is a permutation chosen uniformly at random.

Note that $X \leq 1$. Let's now compute its expected value.

$$\begin{aligned}\mathbb{E}(X) &= \sum_{h \in \mathcal{H}} \sum_{f \in \mathcal{F}} \Pr(h = \pi(f)) \\ &= |\mathcal{H}| \cdot \frac{n}{k} \cdot \frac{1}{\binom{n}{k}}\end{aligned}$$

So since $X \leq 1$, we conclude that

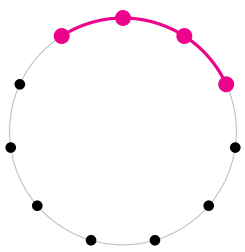
$$|\mathcal{H}| \cdot \frac{n}{k} \cdot \frac{1}{\binom{n}{k}} \leq \mathbb{E}(X) \leq 1$$

Rearranging gives us that

$$|\mathcal{H}| \leq \binom{n}{k} \cdot \frac{k}{n} = \binom{n-1}{k-1}$$

Now let's prove the result in full generality. The random permutation survives, but the partition into n/k blocks does not, and what replaces it is a cycle.

Local Notation. Lay the n vertices out around a circle. A k-**interval** of that cycle is a set of k vertices occupying consecutive positions on it, and $\mathcal{F}_\pi$ will denote the collection of all of them when the vertices are laid out in the order $\pi(1), \pi(2), \ldots, \pi(n)$.



There are n intervals in $\mathcal{F}_\pi$, one starting at each vertex, and—this is the whole point of moving to a cycle—very few of them can pairwise intersect.

Lemma 3.3.3. *Let $n \geq 2k$. Any collection $\mathcal{C} \subseteq \mathcal{F}_\pi$ of pairwise intersecting k-intervals has $|\mathcal{C}| \leq k$. Moreover, if $n \geq 2k+1$ and $|\mathcal{C}| = k$, then $\mathcal{C}$ is the collection of all the k-intervals containing some one vertex.*

Proof. Assume $\mathcal{C}$ is nonempty and fix some $I \in \mathcal{C}$, occupying positions $a, a+1, \ldots, a+k-1$. For $i = 1, \ldots, k-1$, let A_i be the interval ending at position $a+i-1$ and let B_i be the one starting at position $a+i$. Every interval other than I that meets I is one of these $2k-2$: an interval meets I exactly when it starts at one of the $2k-1$ positions $a-k+1, \ldots, a+k-1$, and the A_i take the $k-1$ of those before a while the B_i take the $k-1$ after it. Now A_i and B_i between them occupy the $2k$ consecutive positions $a+i-k, \ldots, a+i+k-1$, which are distinct because $n \geq 2k$, so $A_i \cap B_i = \emptyset$ and $\mathcal{C}$ can contain at most one of the two. Taking I together with one interval from each of the $k-1$ pairs gives $|\mathcal{C}| \leq k$.

Suppose now that $n \geq 2k+1$ and $|\mathcal{C}| = k$, so that $\mathcal{C}$ holds exactly one of A_i, B_i for every i. It cannot hold both A_i and B_{i+1}: those occupy positions $a+i-k, \ldots, a+i-1$ and $a+i+1, \ldots, a+i+k$, which lie among $2k+1$ consecutive positions and so are distinct, leaving the two intervals disjoint. So as soon as some B_j lies in $\mathcal{C}$ so does every B_i with $i < j$, and hence there is an m with

$$\mathcal{C} = \{A_{k-1}, \ldots, A_{m+1}, I, B_1, \ldots, B_m\}$$

Reading off where each of those starts, they are the intervals starting at positions $a+m+1-k, \ldots, a+m$, which is a run of k consecutive positions—and the intervals starting there are precisely the ones containing the vertex at position $a+m$. □

The hypothesis $n \geq 2k+1$ is needed in that second half: on C_6 the three 3-intervals $\{1,2,3\}$, $\{3,4,5\}$ and $\{5,6,1\}$ pairwise intersect, and there are $k = 3$ of them, but no vertex lies in all three.

Proof of the bound in Theorem 3.3.2. Choose $\pi \in S_n$ uniformly at random and set

$$X = |\{h \in \mathcal{H} \mid h \in \mathcal{F}_\pi\}|$$

The intervals being counted here are hyper-edges of H, so they pairwise intersect, and by Lemma 3.3.3

it is always true that $X \leq k$. Now, let's compute the expectation of X.

$$\mathbb{E}(X) = \sum_{h \in \mathcal{H}} \Pr(h \in \mathcal{F}_\pi)$$

A given h is an interval of the randomly ordered cycle exactly when π lays its k elements out consecutively, and we can count the π that do so. There are n places for the interval to start, $k!$ ways of arranging h within it and $(n-k)!$ ways of arranging the other $n-k$ vertices, and no π gets counted twice, since an interval determines where it starts. So

$$\Pr(h \in \mathcal{F}_\pi) = \frac{n \cdot k!\,(n-k)!}{n!} = \frac{n}{\binom{n}{k}}$$

and hence

$$|\mathcal{H}| \cdot \frac{n}{\binom{n}{k}} = \mathbb{E}(X) \leq k$$

Rearranging,

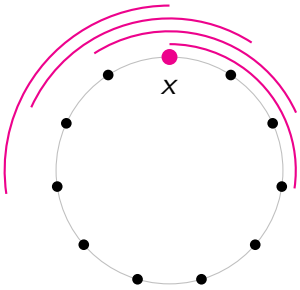
$$|\mathcal{H}| \leq \frac{k}{n}\binom{n}{k} = \binom{n-1}{k-1}$$

□

3.3.2 The Extremal Case

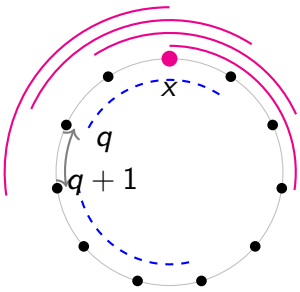
That argument has slack in exactly one place, and at $|\mathcal{H}| = \binom{n-1}{k-1}$ there is none left: X is then equal to k for every ordering rather than merely on average. Once $n \geq 2k+1$, so that the second half of Lemma 3.3.3 is available, every ordering of the cycle hands us a run of k intervals all of which are hyper-edges—and that pins $\mathcal{H}$ down completely. The following is the "moreover" of Theorem 3.3.2, made precise.

Theorem 3.3.4. *Let H be an intersecting k-uniform hypergraph on n vertices. If $|\mathcal{H}| = \binom{n-1}{k-1}$ and $n \geq 2k+1$, then the extremal example is the obvious one: there is a vertex x such that $\mathcal{H}$ consists of precisely the k-subsets of $[n]$ that contain x.*



Whatever order we lay the vertices out in, the picture looks like the one above: the hyper-edges we can see on the cycle are the k intervals through one distinguished vertex, drawn here as concentric arcs. All the work is in showing that the distinguished vertex does not move when the order changes.

Here is the picture to keep in mind. Swapping two neighbours on the cycle disturbs only two of its n intervals—the one ending at the first of them and the one starting at the second, each of which holds exactly one of the pair—and every other interval keeps its vertices. So at least $k-1$ of the k arcs through x are still hyper-edges and still intervals after the swap, and $k-1$ consecutive arcs leave the new fan almost no room to sit anywhere else. The proof below is the bookkeeping that turns "almost" into "none".



Here $n = 11$ and $k = 4$. The two dashed intervals are the ones the swap of positions q and $q+1$ disturbs; one of them is among the four hyper-edges through x and the other is not, and the remaining three arcs are untouched.

Proof of Theorem 3.3.4. Run the argument above again, this time at equality. If $|\mathcal{H}| = \binom{n-1}{k-1}$, then

$$\mathbb{E}(X) = \binom{n-1}{k-1} \cdot \frac{n}{\binom{n}{k}} = k$$

and we already know that $X \leq k$ whatever π is. There are finitely many π and each is equally likely, so a mean of k under a ceiling of k leaves nothing to spare: $X = k$ for every single π. However we lay the n vertices out around a cycle, then, exactly k of that cycle's k-intervals are hyper-edges of H, and by Lemma 3.3.3 those k intervals are the intervals through a single vertex—one vertex and no more, since the outermost two of them share only it. Write x_π for that vertex. For $k = 1$ an intersecting family is a single hyper-edge and the theorem is immediate, so assume from here that $k \geq 2$.

It remains to see that x_π does not depend on π, and one adjacent transposition at a time is enough. Fix an ordering, put $x = x_\pi$ at position 0 and number the other positions $1, \ldots, n-1$ round the circle, and let I_j be the interval occupying *positions* $j, j+1, \ldots, j+k-1$, indices read modulo n. The intervals through position c are $I_{c-k+1}, \ldots, I_c$, so the hyper-edges of this ordering are exactly $I_{-(k-1)}, \ldots, I_0$. Transposing the vertices at adjacent positions q and $q+1$, neither of them position 0, changes which vertices occupy I_j exactly when I_j holds one of those two positions and not the other—which happens for $j = q+1$ and for $j = q-k+1$, and for no other j.

So the new ordering's hyper-edges, again the intervals through a single position c, agree with $I_{-(k-1)}, \ldots, I_0$ outside those two indices. But two runs of k consecutive indices whose right-hand ends sit c apart differ in $2\min(|c|, k)$ places, where $|c|$ is the distance from c to 0 measured around the circle, so $c \neq 0$ would force $|c| = 1$, and the two indices they differ in are then $\{-(k-1), 1\}$ if $c = 1$ and $\{-k, 0\}$ if $c = -1$. Either set would have to be $\{q-k+1, q+1\}$. Both pairs differ by k, and matching them in the only order that does not ask for $-k \equiv k \pmod{n}$ (which is impossible, as $n > 2k$) gives $q = 0$ in the first case and $q = -1$ in the second. Both are excluded, so $c = 0$: the distinguished vertex is the one at position 0, which the transposition left where it was.

Adjacent transpositions avoiding position 0 generate every arrangement of the other $n-1$ vertices around the circle, and each one leaves the distinguished vertex at position 0 for the next to start from, so the paragraph above applies afresh along the whole chain. Rotating or reflecting a layout leaves its intervals, and hence x_π, alone—so every π can first be turned until x sits at position 0, and $x_\pi = x$ for all of them. Take any k-subset h of $[n]$ containing x, and lay the vertices out with the elements of h consecutive: then h is an interval containing x, so $h \in \mathcal{H}$. There are $\binom{n-1}{k-1}$ such h, which is exactly $|\mathcal{H}|$, so they are all of it. □

Chapter 4

Moments and Concentration

The probabilistic method has been a tool in the two chapters before this one—a random colouring for the lower bound on $m(k)$, a random permutation for Sperner's theorem and for Erdős-Ko-Rado, a random set and then a random graph for Theorem 2.2.7—and every one of those arguments has the same shape: compute an expectation, and read a structure off it. This chapter is about what to do when that is not enough. Knowing $\mathbb{E}(X)$ is large says nothing on its own about whether X is ever nonzero, so we bring in the variance, which is enough to settle when a random graph has a triangle and, more generally, when it has a copy of any fixed graph we like. When the variance in turn is too crude—and for the degrees of a random graph it is hopeless—we go past it to the exponential moments, and get a bound that beats every power of $X - \mathbb{E}(X)$ at once.

The last section takes that idea off sums altogether. A quantity that is not a sum of anything—the chromatic number of a random graph, say—concentrates just as sharply provided it can be revealed a little at a time, and the sequence of guesses we make as it is revealed is a martingale.

Everything here is built on Chapter 1, and on Theorem 1.3.1 and Theorem 1.3.3 in particular; nothing else from outside is needed.

4.1 Second Moment Methods

Let X be the number of triangles in the random graph $G_{n,p}$ of Definition 2.2.10. We get

$$\mathbb{E}(X) = \binom{n}{3} p^3$$

If $p = \frac{\omega(1)}{n}$ then $\mathbb{E}(X) \to \infty$, so $G_{n,p}$ is likely to have a triangle.

If $p = o\left(\frac{1}{n}\right)$, then $\mathbb{E}(X) \to 0$, so the probability that $G_{n,p}$ has a triangle goes to 0.

None of this actually tells us whether $G_{n,p}$ actually has a triangle or not. We need something more sophisticated than the expectation, so we turn to the variance, and Corollary 1.3.4 turns the computation into exactly the one we want.

Theorem 4.1.1. *If $np \to \infty$, then with high probability $G_{n,p}$ contains a triangle.*

Proof. For a triple T of vertices write I_T for the indicator of the event that T spans a triangle, so that $X = \sum_T I_T$ and $\mathbb{E}(X) = \binom{n}{3} p^3$. Expanding the variance,

$$\mathrm{Var}(X) = \sum_{T,T'} \left(\mathbb{E}(I_T I_{T'}) - \mathbb{E}(I_T)\,\mathbb{E}(I_{T'}) \right)$$

where the sum runs over ordered pairs of triples. A pair with $|T \cap T'| \leq 1$ contributes nothing: T and T' then have no pair of vertices in common, so I_T and $I_{T'}$ are determined by disjoint sets of edges and are independent. There are at most n^3 pairs with $T = T'$, each contributing $p^3 - p^6 \leq p^3$. And a pair with $|T \cap T'| = 2$ shares exactly one edge, so $T \cup T'$ spans 5 edges and $\mathbb{E}(I_T I_{T'}) = p^5$; there are at most $3n^4$ such pairs, since T' is determined by T, which of the 3 pairs inside T is shared, and the remaining vertex. Hence

$$\mathrm{Var}(X) \leq n^3 p^3 + 3n^4 p^5$$

For $n \geq 6$ we have $\binom{n}{3} \geq \frac{n^3}{12}$, so $\mathbb{E}(X)^2 \geq \frac{n^6 p^6}{144}$ and

$$\frac{\mathrm{Var}(X)}{\mathbb{E}(X)^2} \leq \frac{144}{(np)^3} + \frac{432}{n \cdot np}$$

Both terms tend to 0 when $np \to \infty$, and Corollary 1.3.4 finishes it. □

Can we apply this probabilistic approach in the following more general situation: what is the threshold for a random graph $G_{n,p}$ to contain H as a subgraph, where H is an arbitrary fixed graph?

What controls the answer is not the size of H but the density of its densest part.

Definition 4.1.2 (Maximum Density)**.** Let H be a graph with at least one edge, and for a graph J write $e(J)$ and $v(J)$ for its numbers of edges and of vertices. The **maximum density** of H is

$$\varepsilon(H) := \max_{J \subseteq H,\ v(J)>0} \frac{e(J)}{v(J)}$$

the maximum being over all subgraphs J of H with at least one vertex.

This is some sort of "edge density" or whatever. Think Szemerédi.

Markov's inequality already settles one side of the threshold.

Theorem 4.1.3. *Let H be a fixed graph with at least one edge. If $p = o\left(\frac{1}{n^{\frac{1}{\varepsilon(H)}}}\right)$, then with high probability[a] $G_{n,p}$ contains no H subgraph.*

[a] for instance, the limit of the probability goes to 1 as $n \to \infty$

Proof. Let $J \subseteq H$ attain the maximum in the definition of $\varepsilon(H)$, and write $v = v(J)$ and $e = e(J)$, so that $\frac{1}{\varepsilon(H)} = \frac{v}{e}$. Every copy of H in $G_{n,p}$ contains a copy of J, so it is enough to show that with high probability there is no copy of J.

Let X be the number of copies of J in $G_{n,p}$. A copy is determined by an injection $V(J) \to [n]$, and different injections may give the same copy, so there are at most n^v of them; each is present precisely when its e edges all are, which happens with probability p^e. Hence

$$\mathbb{E}(X) \leq n^v p^e$$

Now $p = o\left(n^{-v/e}\right)$, so $p^e = o(n^{-v})$ and $\mathbb{E}(X) = o(1)$. By Theorem 1.3.1, $\Pr(X \geq 1) \leq \mathbb{E}(X) \to 0$. □

The bound is sharp in the sense that at $p = n^{-1/\varepsilon(H)}$ itself the first moment stops deciding anything, which is where the second moment has to come in.

Example 4.1.4. Let J be K_4 without one edge, so $v(J) = 4$ and $e(J) = 5$. Its subgraphs are sparser than it is—a triangle has density 1, a single edge $\frac{1}{2}$—so $\varepsilon(J) = \frac{5}{4}$ and the threshold sits at $p = \frac{1}{n^{4/5}}$. At exactly that p, each 4-set of vertices carries $\frac{4!}{|\mathrm{Aut}(J)|} = 6$ copies of J, so the number X of copies has

$$\mathbb{E}(X) = \binom{n}{4} \cdot 6 \cdot p^5 = \binom{n}{4} \cdot 6 \cdot \frac{1}{n^4} \to \frac{1}{4}$$

Markov's inequality bounds $\Pr(X \geq 1)$ by about $\frac{1}{4}$ and says nothing at all about whether X is ever positive.

Our goal is to show that once the expected number of Js goes to infinity (J being the densest part of H), we have a copy of H.

4.2 Chernoff's Bound

Consider a random graph $G_{n,p}$ with n vertices and edges chosen uniformly at random with probability p, and take $p = \frac{1}{2}$ for this discussion.

The expected degree of a vertex is $\frac{n-1}{2}$.

How many vertices have degrees lying in $(0.49n, 0.51n)$? We can actually show that

$$\lim_{n\to\infty} \Pr(\text{every vertex in } G_{n,p} \text{ has degree in } (0.49n, 0.51n)) = 1$$

Indeed, fix a vertex x. $\deg(x)$ is a binomial random variable, counting the $n-1$ possible edges at x, each present independently. $\mathbb{E}(\deg(x)) = \frac{n-1}{2}$ and $\mathrm{Var}(\deg(x)) = \frac{n-1}{4}$.

Applying Chebyshev gives us

$$\Pr\left(\left|\deg(x) - \frac{n-1}{2}\right| \geq \delta n\right) \leq \frac{\frac{n-1}{4}}{(\delta n)^2} \approx \frac{1}{4\delta^2 n}$$

But this is a rather terrible bound: for $\delta = 0.01$, this is already $\frac{2{,}500}{n}$. Saying a probability is less than or equal to this is essentially useless!

We can consider taking higher moment bounds: bounding using not the second power of $X - \mathbb{E}(X)$, but the fourth or sixth or eighth power. In this instance, a fourth power bound would actually be enough. But that won't always be the case.

What's one thing that beats *all* of these? An exponential!

Consider the following setup. Let $\xi_1, \ldots, \xi_n$ be independent Bernoulli random variables take values ± 1 with probability $\frac{1}{2}$. Define the random variable $\eta := \sum_{i=1}^{n} \xi_i$.

Note that because $x < y \iff e^x < e^y$, we get that

$$\Pr(\eta \geq a) = \Pr(\beta^\eta \geq \beta^a) \leq \frac{\mathbb{E}(\beta^\eta)}{\beta^a}$$

by Markov's inequality (where β denotes e^t for $t > 0$, a bit like how q denotes $e^{2\pi i z}$).

Optimising over t turns that into a bound that decays in a faster than any power.

Theorem 4.2.1 (Chernoff's Bound for Probability $\frac{1}{2}$). *Let $\xi_1, \ldots, \xi_n$ be independent random variables each taking the values ± 1 with probability $\frac{1}{2}$, and let $\eta = \sum_{i=1}^{n} \xi_i$. Then for every $a \geq 0$,*

$$\Pr(\eta \geq a) \leq e^{-\frac{a^2}{2n}}$$

Proof. At $a = 0$ the bound reads $\Pr(\eta \geq 0) \leq 1$ and there is nothing to prove, so take $a > 0$, and fix $t > 0$ for the moment. Notice that

$$\mathbb{E}(e^{t\eta}) = \mathbb{E}\left(e^{t\sum_{i=1}^{n}\xi_i}\right) = \mathbb{E}\left(\prod_{i=1}^{n} e^{t\xi_i}\right) = \prod_{i=1}^{n} \mathbb{E}\left(e^{t\xi_i}\right) = \left(\frac{e^{-t} + e^t}{2}\right)^n \tag{4.2.1}$$

the third equality by independence. Comparing the series termwise, $\frac{e^t + e^{-t}}{2} = \sum_{k \geq 0} \frac{t^{2k}}{(2k)!} \leq \sum_{k \geq 0} \frac{t^{2k}}{2^k k!} = e^{\frac{t^2}{2}}$, since $(2k)! \geq 2^k k!$. We therefore conclude that

$$\Pr(\eta \geq a) = \Pr(e^{t\eta} \geq e^{ta}) \leq \frac{\mathbb{E}(e^{t\eta})}{e^{ta}} \leq e^{\frac{t^2 n}{2} - ta}$$

As this result holds for all t, can we optimise this? Ie, can we find the t that minimises this? It turns out, we can: by some calculus nonsense, we get that $t = \frac{a}{n}$ is optimal, giving us the upper-bound $e^{-\frac{a^2}{2n}}$. □

The same bound holds if the ξ_i are independent and for all i, $|\xi_i| \leq 1$ and $\mathbb{E}(\xi_i) = 0$.

For instance, consider the case where

$$\xi_i = \begin{cases} \frac{1}{3} & \text{with probability } \frac{3}{4} \\ -1 & \text{with probability } \frac{1}{4} \end{cases}$$

and as before, let η denote their sum. By the same sort of technology we used above,

$$\mathbb{E}(e^{t\eta}) = \prod_{i=1}^{n} \mathbb{E}\left(e^{t\xi_i}\right)$$

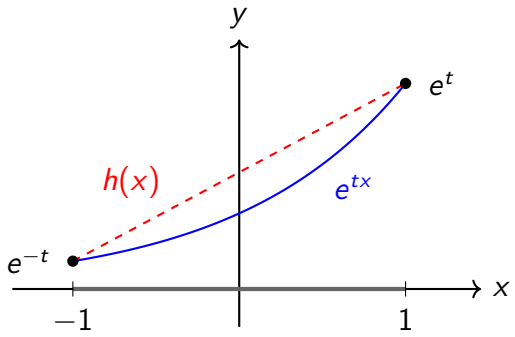
Define

$$h(x) = \frac{e^t + e^{-t}}{2} + x\frac{e^t - e^{-t}}{2}$$

We can show that

$$\mathbb{E}\left(e^{t\xi_i}\right) \leq \mathbb{E}(h(\xi_i)) = h(\mathbb{E}(\xi_i)) = h(0) = \frac{e^t + e^{-t}}{2} \tag{4.2.2}$$

The first inequality is what makes this work: h is the chord of $x \mapsto e^{tx}$ across $[-1, 1]$, and since e^{tx} is convex it lies below that chord on the whole interval, which is where $|\xi_i| \leq 1$ is used. The first equality is linearity, since h is affine, and the second is where $\mathbb{E}(\xi_i) = 0$ is used.



We thus get, more generally, the following result.

Theorem 4.2.2 (Chernoff's Bound). *Let $\xi_1, \ldots, \xi_n$ be independent random variables with $|\xi_i| \leq 1$ and $\mathbb{E}(\xi_i) = 0$ for all i. Let $\eta = \sum_{i=1}^{n} \xi_i$. Then for every $a \geq 0$,*

$$\Pr(\eta \geq a) \leq e^{-\frac{a^2}{2n}}$$

Proof sketch. We prove this identically to Theorem 4.2.1, except in our chain of equalities (4.2.1), we replace the last equality with a $\leq$, as given by (4.2.2). □

4.3 Martingales

Theorem 4.2.2 got its strength from a very particular structure: a sum of independent contributions, each of them small. Plenty of quantities we care about have no such structure—the chromatic number of a random graph is not a sum of anything—and concentrate just as sharply anyway. What they do have is that they can be revealed a little at a time, with each revelation moving the expectation only slightly, and that is enough.

Throughout this section, fix random variables $X_0, X_1, X_2, \ldots$ on some probability space $(\Omega, \mathcal{P}(\Omega), \Pr)$.

The sequence to have in mind is the one where X_k is our best guess at some quantity after k things have been revealed, so that each term is the average of the term after it.

Definition 4.3.1 (Martingale)**.** We say $X_0, X_1, \ldots$ is a **martingale** if the following all hold:

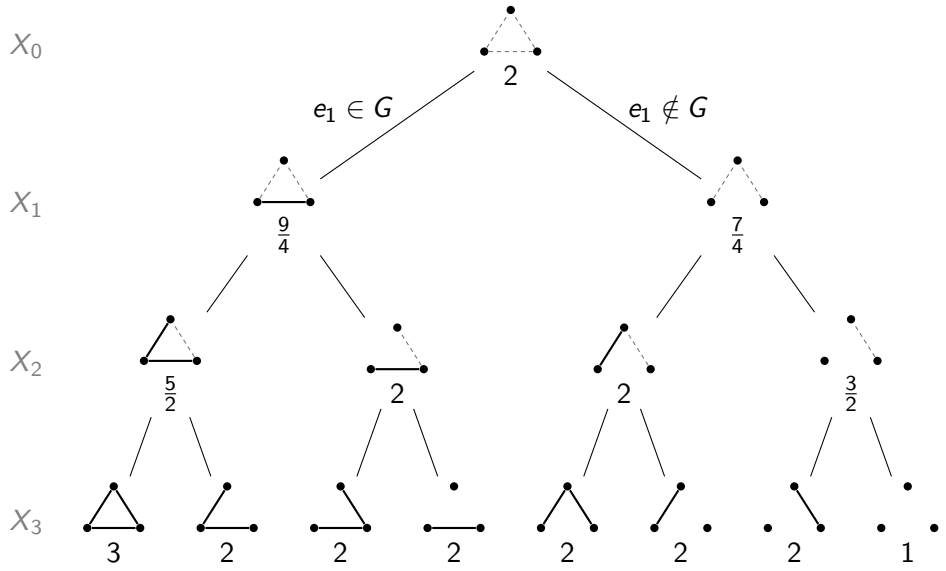
1. for all integers $n \geq 0$, $|X_{n+1} - X_n| \leq 1$
2. the condition below is imposed only where $\Pr(X_0 = x_0, \ldots, X_n = x_n) \neq 0$, the conditional expectation being undefined otherwise
3. for all integers $n \geq 0$ and all $x_0, \ldots, x_n \in \mathbb{R}$,

$$\mathbb{E}(X_{n+1} \mid X_0 = x_0, X_1 = x_1, \cdots, X_n = x_n) = x_n$$

4.3.1 Exposing a Random Graph

Let's study the expected value of chromatic numbers (cf. Definition 2.2.1). Consider the family of random graphs $G_{3,\frac{1}{2}}$ of graphs on 3 vertices with the probability of an edge between them being $\frac{1}{2}$ (cf. Definition 2.2.10). This makes $\chi\left(G_{3,\frac{1}{2}}\right)$ a random variable. What would we expect its expectation to be?

We can heuristically see that the expected chromatic number is two based on the following figure:



Fix an order e_1, e_2, e_3 on the three possible edges and look at them one at a time. An edge not yet looked at is drawn dashed, since it is there with probability $\frac{1}{2}$; one already looked at is drawn solid if it is there and rubbed out if it is not. So the root has all three dashed, its two children split on e_1, theirs on e_2, and the eight leaves are the eight graphs on three vertices.

Under each picture is the expected chromatic number of a graph from the family it stands for—at a leaf, just that graph's: 1 for the empty graph, 2 for one or two edges, 3 for the triangle. Each is the average of the two below it, so the 2 at the root is $\mathbb{E}\left(\chi\left(G_{3,\frac{1}{2}}\right)\right)$.

Write G for $G_{3,\frac{1}{2}}$ and $e_1, \ldots, e_m$ for the $m = \binom{3}{2}$ possible edges.

Define the following martingale:

$$
\begin{aligned}
X &= \chi(G) \\
Y_1 &= \mathbf{1}_{e_1 \in G} \\
&\vdots \\
Y_m &= \mathbf{1}_{e_m \in G} \\
X_0 &= \mathbb{E}(X) \\
X_1 &= \mathbb{E}(X \mid Y_1) \\
X_2 &= \mathbb{E}(X \mid Y_1, Y_2)
\end{aligned}
$$

⋮

Then, $X_0, X_1, X_2, \ldots$ is a martingale, and it is exactly what the tree draws: X_k is the number written under the picture at the node reached by exposing $e_1, \ldots, e_k$. Its increments are never more than $\frac{1}{2}$, so the first condition is met with room to spare; X_0 is the 2 at the root, and $X_m = X$ is the chromatic number itself, every X_k past that being X over again with nothing left to expose.

We can adapt this to the general $G_{n,\frac{1}{2}}$ case. Define Y_i and X_i as above. Then, we can compute the conditional expectation of $\chi\left(G_{n,\frac{1}{2}}\right)$ on observing edges incident on vertices $1, \ldots, k$: write X for $\chi\left(G_{n,\frac{1}{2}}\right)$, let Y_k record the edges from vertex k back to $1, \ldots, k-1$, and set $X_k = \mathbb{E}(X \mid Y_1, \ldots, Y_k)$. Since Y_1 records nothing, $X_0 = X_1 = \mathbb{E}(X)$; and $X_n = X$, with $X_k = X$ beyond that as before.

We are now exposing a vertex at a time rather than an edge at a time, which is what bounds the increments. Two graphs differing only at one vertex have chromatic numbers differing by at most 1: give that vertex a colour of its own. Pairing each unexposed configuration with itself under a different value of Y_{k+1}, the possible values of X_{k+1} differ pairwise by at most 1, and X_k is their average, so $|X_{k+1} - X_k| \leq 1$.

4.3.2 Azuma's Inequality

That condition is what gives us the following: a martingale cannot get far from where it started, and by the same margin Theorem 4.2.2 gives a ± 1 walk.

Theorem 4.3.2 (Azuma's Inequality). *Let $X_0, X_1, \ldots$ be a martingale and let m be a positive integer. Then for every $\lambda > 0$,*

$$\Pr\left(X_m - X_0 \geq \lambda\sqrt{m}\right) \leq e^{-\frac{\lambda^2}{2}}$$

We won't prove it: it is the argument behind Theorem 4.2.2, with the chord bound applied to each increment conditionally on what has been exposed—which is what the bound on the increments is there to license.

So $\chi\left(G_{n,\frac{1}{2}}\right)$ sits in a window of width about $\sqrt{n}$ around its mean—much narrower than χ itself,

which grows like $\frac{n}{\log n}$ (not something we will prove).

Corollary 4.3.3. *For every $n \geq 1$ and every $\lambda > 0$,*

$$\Pr\left(\left|\chi\left(G_{n,\frac{1}{2}}\right) - \mathbb{E}\left(\chi\left(G_{n,\frac{1}{2}}\right)\right)\right| \geq \lambda\sqrt{n}\right) \leq 2e^{-\frac{\lambda^2}{2}}$$

Proof. The martingale above has $X_0 = \mathbb{E}\left(\chi\left(G_{n,\frac{1}{2}}\right)\right)$ and $X_n = \chi\left(G_{n,\frac{1}{2}}\right)$, so Theorem 4.3.2 at $m = n$ bounds $\Pr\left(X_n - X_0 \geq \lambda\sqrt{n}\right)$ by $e^{-\frac{\lambda^2}{2}}$. Applying it to $-X_0, -X_1, \ldots$, a martingale for the same reasons, bounds the other tail. □

4.3.3 The Doob Martingale

It turns out that the martingale we constructed for the chromatic number problem is a specific instance of a more general martingale, known as the Doob Martingale.

Definition 4.3.4 (Doob Martingale)**.** For random variables $X, Y_1, Y_2, \ldots$ on some probability space, define

$$X_0 := \mathbb{E}(X) \qquad \forall i \geq 1,\ X_i := \mathbb{E}(X \mid Y_1, \ldots, Y_i)$$

It is possible to show that $X_0, X_1, X_2, \ldots$ always satisfies the averaging condition of Definition 4.3.1; once its increments are bounded by 1 it is therefore a martingale, called the **Doob Martingale**.

4.3.4 McDiarmid's Inequality

What makes Definition 4.3.4 worth having is that its increments are controlled as soon as the Y_i are independent and X depends on no one of them too heavily.

Definition 4.3.5 (Coordinatewise Lipschitz)**.** Given metric spaces $(X_i, d_i)_{i=1}^n$ and (Y, u), we say a function $f : X_1 \times \cdots \times X_n \to Y$ is **k-coordinatewise-Lipschitz** if $u(f(\mathbf{a}), f(\mathbf{b})) \leq k$ if $\mathbf{a}$ and $\mathbf{b}$ differ in exactly one coordinate.

Such an f, fed through the Doob martingale of its arguments, concentrates as sharply as a sum

would.

Proposition 4.3.6 (McDiarmid's Inequality). *Suppose $Y_1, \ldots, Y_n$ are independent random variables on domains $\mathcal{Y}_1, \ldots, \mathcal{Y}_n$. Suppose $f : \mathcal{Y}_1 \times \cdots \times \mathcal{Y}_n \to \mathbb{R}$ is 1-coordinatewise-Lipschitz. Denoting $X = f(Y_1, \ldots, Y_n)$, we have that*

$$\Pr(X \geq \mathbb{E}(X) + a) \leq e^{-\frac{a^2}{2n}}$$

for all $a \geq 0$.

Proof. Consider the Doob Martingale $X_i = \mathbb{E}(X \mid Y_1, \ldots, Y_i)$. Suppose we've fixed $y_1, \ldots, y_{i-1}$. Define

$$g_i(y) = \mathbb{E}(f(y_1, \ldots, y_{i-1}, y, Y_{i+1}, \ldots, Y_n))$$

sorry □

One application of Proposition 4.3.6 is something called the Smiley Face Lemma, from a paper of Alan Frieze and Wes [1].

The lemma counts copies of a pattern that are both accurate and isolated.

Definition 4.3.7 ((ε, D)-Copy). Let $S \subseteq \mathbb{R}^2$ be finite, $\varepsilon > 0$ and $D > 0$. An (ε, D)-**copy** of S in a finite set $X \subseteq \mathbb{R}^2$ is a subset $T \subseteq X$ for which there is a bijection f from T to a set S' congruent to S with $\|x - f(x)\| < \varepsilon$ for every $x \in T$, and such that every point of T is at distance more than D from every point of $X \setminus T$.

Scattering n points at random over a square of area n leaves linearly many such copies.

Lemma 4.3.8 (Smiley Face Lemma, [1, Observation 3.1]). *For any fixed S, ε and D as above, there is some $\delta > 0$ such that n uniformly random points in a $\sqrt{n} \times \sqrt{n}$ square contain at least δn many (ε, D)-copies of S, with high probability.*

Chapter 5

The Local Lemma

Whenever we have wanted a random object to avoid a list of bad events, we have asked for the bad events to be rare enough that, added up, they still cannot cover everything—the union bound behind Theorem 2.1.10. This chapter asks for much less: there may be far too many bad events for that, provided each one depends on only a few of the others.

5.1 The Lovász Local Lemma

Suppose $B_1, \ldots, B_n$ are independent events with $\Pr(B_i) < 1$. It's pretty easy to see that $\Pr\left(\bigcap B_i^C\right) > 0$. Essentially, we think of the B_i as *bad events*: events we *don't* want to see happen.

We want to extend this. In what way exactly? Here's a motivating question.

In a k-uniform hypergraph where each edge intersects at most D others, what condition on k makes the hypergraph 2-colourable?

What we'd want to do is randomly colour vertices: each edge f is monochromatic with probability $\Pr(B_f) = \frac{1}{2^{k-1}}$. We would then like to apply some version of the fact that $\Pr\left(\bigcap B_i^C\right) > 0$.

Two edges sharing a vertex are plainly not independent, so what we ask instead is that each bad event depend on only a few of the others—and a graph is the natural place to record which.

Definition 5.1.1 (Dependency Graph). Given a collection $\mathcal{A}$ of events, a **dependency graph on** $\mathcal{A}$ is a graph with vertices $\{v_A \mid A \in \mathcal{A}\}$ such that $v_A \not\sim v_{B_1}, \ldots, v_{B_n}$ implies that A is independent of the family $B_1, \ldots, B_n$ (in the sense of Definition 1.1.1).

Another way of phrasing this is that dependence forces adjacency: a dependency graph may join more than it has to, but it may not leave a dependence unrecorded.

Local Notation (Dependence). Fix a dependency graph on $\mathcal{A}$. Borrowing from our use of $\sim$ for adjacency, we will write $A \sim B$ for $v_A \sim v_B$, and read it as saying that A and B are permitted to depend on each other.

The extension we were after weighs each event against the ones it depends on.

Theorem 5.1.2 (Asymmetric Lovász Local Lemma). *Given a finite collection $\mathcal{A}$ of events and real numbers $0 < x_A < 1$ such that for all $A \in \mathcal{A}$,*

$$\Pr(A) \leq x_A \prod_{B \sim A} (1 - x_B)$$

we can conclude that

$$\Pr\left(\bigcap_{A \in \mathcal{A}} A^C\right) > 0$$

Proof. We show that for every $A \in \mathcal{A}$ and every $S \subseteq \mathcal{A} \setminus \{A\}$,

$$\Pr\left(A \,\middle|\, \bigcap_{B \in S} B^C\right) \leq x_A$$

by induction on $|S|$. (Every conditional probability below is defined: the last display of the proof, applied to S in place of $\mathcal{A}$, gives $\Pr\left(\bigcap_{B \in S} B^C\right) \geq \prod_{B \in S} (1 - x_B) > 0$, and only uses the claim for smaller sets.) When $S = \emptyset$ this is the hypothesis, as $\prod_{B \sim A} (1 - x_B) \leq 1$. Otherwise split S into $S_1 = \{B \in S \mid B \sim A\}$ and $S_2 = S \setminus S_1$, and write $E_i = \bigcap_{B \in S_i} B^C$, so that

$$\Pr(A \mid E_1 \cap E_2) = \frac{\Pr(A \cap E_1 \mid E_2)}{\Pr(E_1 \mid E_2)}$$

The numerator is at most $\Pr(A \mid E_2)$, and E_2 is built out of events not adjacent to A, so this is

$\Pr(A) \le x_A \prod_{B\sim A}(1-x_B)$. For the denominator, enumerate $S_1 = \{B_1,\dots,B_r\}$ and write

$$\Pr(E_1 \mid E_2) = \prod_{i=1}^{r} \Pr\big(B_i^C \mid B_1^C \cap \cdots \cap B_{i-1}^C \cap E_2\big) \ge \prod_{i=1}^{r}(1-x_{B_i}) \ge \prod_{B\sim A}(1-x_B)$$

where each factor is bounded by the induction hypothesis, since each conditions on fewer than $|S|$ events. Dividing, $\Pr(A \mid E_1 \cap E_2) \le x_A$.

Finally, enumerate $\mathcal{A} = \{A_1,\dots,A_n\}$, so that

$$\Pr\left(\bigcap_{i=1}^{n} A_i^C\right) = \prod_{i=1}^{n} \Pr\big(A_i^C \mid A_1^C \cap \cdots \cap A_{i-1}^C\big) \ge \prod_{i=1}^{n}(1-x_{A_i}) > 0$$

□

In applications the bad events usually all have about the same probability and about the same number of neighbours, and then one choice of weights does for all of them.

Corollary 5.1.3 (Symmetric Lovász Local Lemma). *Let $\mathcal{A}$ be a finite collection of events with a dependency graph in which every vertex has at most d neighbours, and suppose that $\Pr(A) \le p$ for all $A \in \mathcal{A}$. If*

$$ep\,(d+1) \le 1$$

then $\Pr\big(\bigcap_{A\in\mathcal{A}} A^C\big) > 0$.

Proof. If $d < 1$ the graph has no edges, so each $A \in \mathcal{A}$ is independent of the family of all the others, and $\Pr\big(\bigcap_A A^C\big) = \prod_A (1-\Pr(A)) \ge (1-p)^{|\mathcal{A}|} > 0$, as $p \le \frac{1}{e}$. Otherwise take $x_A = \frac{1}{d+1} \le \frac{1}{2}$ for every A. Then for every $A \in \mathcal{A}$,

$$x_A \prod_{B\sim A}(1-x_B) \ge \frac{1}{d+1}\left(1-\frac{1}{d+1}\right)^{d} \ge \frac{1}{e\,(d+1)} \ge p \ge \Pr(A)$$

using $\left(1+\frac{1}{d}\right)^{d} \le e$ for the second inequality, so Theorem 5.1.2 applies. □

5.2 Applications of the Local Lemma

5.2.1 Binary Sequences

The first application of the local lemma we have in view is to binary sequences, and apart from the lemma itself its argument will need topology (cf. Lemma 2.3.3).

Theorem 5.2.1. *For all $\varepsilon > 0$, there exists some N_ε and an infinite binary sequence in which any two identical blocks of length $n > N_\varepsilon$ are separated by distance at least $(2-\varepsilon)^n$.*

More applications will be added here.

Chapter 6

Ramsey Theory

Turning the colouring question of Chapter 2 around—how large must a complete graph be before every colouring of its edges is forced to produce something monochromatic—gives us the Ramsey numbers, of graphs and then of hypergraphs, and this chapter is about that question and the family of results it has grown into. After the Ramsey numbers themselves we ask it of structures that are not graphs at all, and what is forced is no longer a clique: in the integers it is an arithmetic progression, which is van der Waerden's theorem, and in the hypercube $[n]^d$ it is a combinatorial line, which is the Hales–Jewett theorem. Both are proved by the same device, colour-focusing, and the second contains the first and a good deal more besides. We close with the complete graph on $\mathbb{N}$ itself, where Ramsey's theorem is easier than in the finite case and where one of its finite consequences turns out to lie beyond Peano arithmetic.

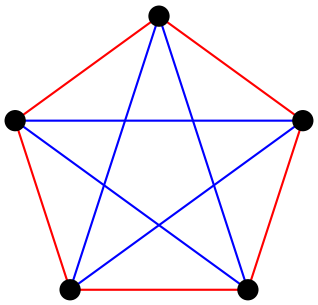
6.1 Ramsey Numbers

Chapter 2 asked when a colouring exists. The opposite question is at least as natural: how large does a structure have to be before every colouring of it is forced to contain something monochromatic? We ask it first of the complete graphs, colouring edges this time rather than vertices, and then of hypergraphs.

6.1.1 Ramsey Numbers of Graphs

Definition 6.1.1 (Ramsey Number). The kth Ramsey number, denoted $R(k)$, is the smallest $n \in \mathbb{N}$ such that any 2-colouring of edges of K_n gives a monochromatic K_k.

One immediate result we can see is that $R(3) \geq 6$, which we can see simply by drawing a picture of K_5 and studying what happens if we move our attention to K_6.



Each colour carries a 5-cycle—the outer pentagon in red, the inner pentagram in blue—and a 5-cycle contains no triangle at all, so neither colour gives us a monochromatic K_3. Nothing forces a monochromatic triangle at $n = 5$, and so $R(3) \geq 6$. The picture also shows how little room there is to spare: every vertex of K_5 meets exactly two edges of each colour, and the moment we pass to K_6 each vertex has five edges to distribute between two colours, so one of the colours is forced to take three of them.

We can actually prove that $R(3) = 6$, but this is rather nontrivial.

Other known Ramsey numbers are $R(2) = 2$ and $R(4) = 18$. The rest are unknown, though we do have bounds for them. The best (or close to the best) known bounds are

$$2^{\frac{k}{2}} \leq R(k) \leq 3.8^k$$

Upper bounds of this kind come from a recursion, which is cleanest stated for the off-diagonal Ramsey number $R(k, \ell)$: the smallest $n \in \mathbb{N}$ such that any red/blue colouring of the edges of K_n gives a red K_k or a blue K_ℓ, so that $R(k) = R(k, k)$.

Theorem 6.1.2. *For all $k, \ell \geq 2$, we have*

$$R(k, \ell) \leq R(k-1, \ell) + R(k, \ell-1)$$

See undergraduate graph theory notes.

add cross-ref to Michele's notes

6.1.2 Hypergraph Ramsey Numbers

Nothing in the question needs the things being coloured to be the edges of a graph, and our motivation for going further is the following result about sequences, which we state without proof.

Theorem 6.1.3 (Erdős-Szekeres). *If $n \geq (r-1)(s-1)+1$, then any real sequence $a_1, \ldots, a_n$ has a non-decreasing subsequence of length r or a decreasing subsequence of length s.*

To ask the question of hypergraphs we need the hypergraph with every possible hyper-edge.

Definition 6.1.4 (Complete Uniform Hypergraph). Define the **complete r-uniform hypergraph on n vertices**, denoted $K_{(r)}(n)$, to be the r-hypergraph on n vertices with all possible hyper-edges.

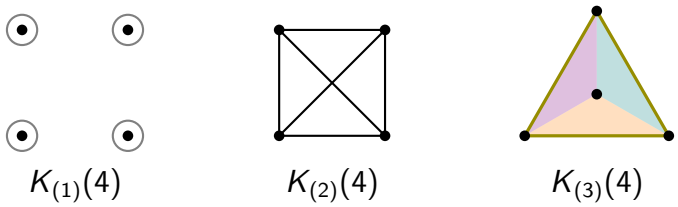
What we look for inside it is the analogue of a monochromatic K_k.

Definition 6.1.5 (k-clique). Given a hypergraph and a colouring of its hyper-edges, a **k-clique of some colour** is a set of k vertices such that all possible hyper-edges on that vertex set are monochromatic, of that colour. The **size** of a k-clique is its number of vertices, k.

The Ramsey numbers of graphs then generalise directly.

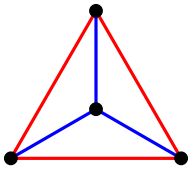
Definition 6.1.6 (Hypergraph Ramsey Number). Given $r, k, \ell \in \mathbb{N}$, we define $R_{(r)}(k, \ell)$ to be the least n such that any red/blue colouring of the hyper-edges of $K_{(r)}(n)$ contains a red k-clique or a blue ℓ-clique.

For small r these are familiar objects. $K_{(2)}(n)$ is just K_n, so $R_{(2)}(k,k)$ is the Ramsey number $R(k)$ above, and in $K_{(1)}(n)$ each vertex is a hyper-edge by itself. Below are $K_{(1)}(4)$, $K_{(2)}(4)$ and $K_{(3)}(4)$, the last with its four hyper-edges drawn as triangles: three shaded ones, and the outline of the fourth around them.



$K_{(1)}(4)$ $K_{(2)}(4)$ $K_{(3)}(4)$

Once the hyper-edges are coloured, a clique of a colour is a set of vertices all of whose hyper-edges have that colour: in the colouring of $K_{(2)}(4)$ below, the red triangle is a red 3-clique, and each blue edge is a blue 2-clique.



One easy fact to see is the following.

Lemma 6.1.7. $R_{(1)}(k,\ell) = k + \ell - 1$.

Proof. In $K_{(1)}(n)$ every hyper-edge is a single vertex, so colouring the hyper-edges colours the vertices, and a red k-clique is just k red vertices. If $n \geq k + \ell - 1$, then fewer than k red vertices leaves at least ℓ blue ones. If $n \leq k + \ell - 2$, colouring at most $k - 1$ vertices red and at most $\ell - 1$ blue gives neither. □

6.1.3 A Recursive Upper Bound

We can compute a similar bound to what we know about Ramsey numbers on graphs (Theorem 6.1.2).

Theorem 6.1.8. *For all $r, k, \ell \geq 2$, we have*

$$R_{(r)}(k,\ell) \leq R_{(r-1)}\big(R_{(r)}(k-1,\ell)\,, R_{(r)}(k,\ell-1)\big) + 1$$

Proof. Write $A = R_{(r)}(k-1,\ell)$ and $B = R_{(r)}(k,\ell-1)$, and take $n = R_{(r-1)}(A,B)+1$.

Consider $K_{(r)}(n)$, the complete r-uniform hypergraph on $[n]$. Consider any red/blue colouring of its edges. Fix $v \in V\big(K_{(r)}(n)\big)$. There are $R_{(r-1)}(A,B)$ other vertices.

Denote by H the $(r-1)$-uniform hypergraph on those vertices made by removing v from every edge containing v. Every $(r-1)$-subset S of those vertices is the shadow $(S\cup\{v\})\setminus\{v\}$ of exactly one edge, so H is a copy of $K_{(r-1)}(n-1)$. Define a "shadow colouring" of H by letting the colour of the "shadow edge" $f\setminus\{v\}$ be just the colour of f.

By induction/definition of $R_{(r-1)}$, the shadow contains either a red clique of size A or a blue clique of size B.

Case A: Shadow colouring has a red clique of size A. Let W be its vertex set, so that $|W| = A = R_{(r)}(k-1,\ell)$ and every r-set $S\cup\{v\}$ with S an $(r-1)$-subset of W is red. The original colouring of the r-subsets of W is a red/blue colouring of a copy of $K_{(r)}(A)$, so W contains either a blue ℓ-clique, and we are done, or a red $(k-1)$-clique U. In the second case $U\cup\{v\}$ is a red k-clique: an r-subset of it either lies inside U, and is red because U is a red clique, or is $S\cup\{v\}$ for an $(r-1)$-subset S of $U\subseteq W$, and is red because W is a red clique of the shadow.

Case B: Shadow colouring has a blue clique of size B. The same argument with the colours exchanged: its vertex set has $B = R_{(r)}(k,\ell-1)$ vertices, so it contains either a red k-clique, and we are done, or a blue $(\ell-1)$-clique U, and then $U\cup\{v\}$ is a blue ℓ-clique.

Either way, the colouring has a red k-clique or a blue ℓ-clique, so $R_{(r)}(k,\ell)\leq n = R_{(r-1)}(A,B)+1$. □

At $r=2$ the bound, together with Lemma 6.1.7, reads $R(k,\ell)\leq R(k-1,\ell)+R(k,\ell-1)$, which is Theorem 6.1.2. More importantly, the recursion is what shows that these numbers exist at all:

the definition asks for a least n, and nothing so far has produced even one.

Corollary 6.1.9. *For all $r, k, \ell \in \mathbb{N}$, the Ramsey number $R_{(r)}(k, \ell)$ exists.*

Proof. Induct on r. The case $r = 1$ is Lemma 6.1.7. For $r \geq 2$, induct on $k + \ell$. If $k = 1$ or $\ell = 1$, then $R_{(r)}(k, \ell) = 1$, since a single vertex has no r-subsets and so is vacuously a clique of both colours. Otherwise $k, \ell \geq 2$, so $R_{(r)}(k-1, \ell)$ and $R_{(r)}(k, \ell-1)$ exist by the induction on $k + \ell$, $R_{(r-1)}$ of them exists by the induction on r, and the proof of Theorem 6.1.8 exhibits an n with the required property. □

6.1.4 A Better Upper Bound

Theorem 6.1.8 gives us a recursive bound on hypergraph Ramsey numbers:

$$R_{(r)}(k, \ell) \leq R_{(r-1)}\big(R_{(r)}(k-1, \ell), R_{(r)}(k, \ell-1)\big) + 1$$

Recall also that

$$R_{(2)}(k, \ell) \leq \binom{k + \ell - 2}{k - 1} \leq 2^{k+\ell}$$

We can apply this to show, for $r = 3$, that

$$\begin{aligned} R_{(3)}(k, \ell) &\leq R_{(2)}\big(R_{(3)}(k-1, \ell), R_{(3)}(k, \ell-1)\big) + 1 \\ &\leq 2^{R_{(3)}(k-1,\ell) + R_{(3)}(k,\ell-1)} \\ &\leq \left.2^{2^{\cdot^{\cdot^{\cdot^{2}}}}}\right\} \text{roughly } k + \ell \end{aligned}$$

We can apply this to bound $R_{(4)}$ as well:

$$\begin{aligned} R_{(4)}(k, \ell) &\leq R_{(3)}\big(R_{(4)}(k-1, \ell), R_{(4)}(k, \ell-1)\big) + 1 \\ &\leq \left.2^{2^{\cdot^{\cdot^{\cdot^{2}}}}}\right\} \text{roughly } R_{(4)}(k-1, \ell) + R_{(4)}(k, \ell-1) \\ &\leq \underbrace{\left.2^{2^{\cdot^{\cdot^{\cdot^{2}}}}}\right\} \left.2^{2^{\cdot^{\cdot^{\cdot^{2}}}}}\right\} \cdots \left.2^{2^{\cdot^{\cdot^{\cdot^{2}}}}}\right\} 2}_{k+\ell \text{ towers}} \end{aligned}$$

This is a "super-tower" of depth $k + \ell$. As you can imagine, doing the same for $R_{(5)}$ gives a "super-super-tower", and the upper-bound gets incredibly large incredibly quickly. What this tells us is that we need to improve the upper-bound substantially if we want any hope of estimating hypergraph Ramsey numbers computationally.

We will do two things:

1. Improve the bound on $R_{(r)}(k, \ell)$

2. Improve the bound even further when we're dealing with convex problems

Our strategy will be to find an n large enough so that any red/blue colouring of $K_{(r)}(n)$ admits a red k-clique or a blue ℓ-clique. This n will serve as our upper-bound for $R_{(r)}(k, \ell)$. We will do this by constructing a sequence that contains either a $(k-1)$-clique or an $(\ell-1)$-clique that can be extended to a k-clique or an ℓ-clique by the addition of another element.

Let's begin by improving the upper-bound on $R_{(3)}(k, \ell)$. We'll do this by constructing a sequence $x_1, x_2, \ldots, x_m, x_{m+1}$ of vertices of our hypergraph such that the colour of $\{x_a, x_b, x_c\}$ depends only on a and b for any $a < b < c$. In other words, the colour of x_c should be a *completion* of the colouring of x_a and x_b.[1]

Pick some x_1 and x_2. At least half of the remaining vertices complete it in a consistent colour[2]. *Throw away the rest.* Pick an x_3 from what remains. To pick x_4, repeat this process by considering completions of the pair x_1, x_3 as well as the pair x_2, x_3 (we're already in the $\sim$half consisting of things that consistently complete x_1, x_2). Keep going. Indeed, the number of remaining vertices gets very small very quickly. If you start off with (roughly) $n = 2^{\binom{m}{2}}$ (or maybe $n = 2^{\binom{m}{2}}$ + a small number) vertices in your hypergraph, you can choose $x_1, \ldots, x_m$ via this process, and set x_{m+1} to be what's left. We need to now choose m so that this induces a red k-clique or a blue ℓ-clique, and this gives us our Ramsey upper-bound n.

Suppose $m = R_{(2)}(k-1, \ell-1)$. Then in the colouring induced on pairs in our sequence, we can find a red $(k-1)$-clique or a blue $(\ell-1)$-clique. Adding x_{m+1} then gives us either a red k (3)-clique

[1] a, b, c may not be consecutive: in fact, we will often have rather sparse sequences in the hypergraph.
[2] ie, all of the remaining vertices complete it *in the same way/with the same colour*

or a blue ℓ (3)-clique. This gives us an upper-bound

$$R_{(3)}(k,\ell) \leq 2^{\binom{m}{2}+1} \leq 2^{m^2}$$

(the $+1$ takes care of the "small number" from before). Since $m \leq 2^{k+\ell}$, we can upper-bound the entire thing by

$$R_{(3)}(k,\ell) \leq 2^{2^{2(k+\ell)}}$$

Now let's compute the bound on $R_{(r)}(k,\ell)$.

Theorem 6.1.10. *For all r, k, ℓ sufficiently large, we have the upper-bound*

$$R_{(r)}(k,\ell) \leq 2^{\binom{m}{r-1}+1} \leq 2^{m^{r-1}} = 2^{R_{(r-1)}(k-1,\ell-1)^{r-1}}$$

where $m = R_{(r-1)}(k-1,\ell-1)$.

Proof. We'll proceed in an identical fashion: Given a red/blue colouring of $K_{(r)}(n)$, build a sequence $x_1, \ldots, x_m, x_{m+1}$ such that the colour of any $\{x_{i_1}, \ldots, x_{i_r}\}$ (for $i_1 < i_2 < \cdots < i_r$) is determined by $x_{i_1}, \ldots, x_{i_{r-1}}$. For $n = 2^{\binom{m}{r-1}}$, we can make this sequence.

It remains to choose m so that this sequence has the clique-property. Take $m = R_{(r-1)}(k-1,\ell-1)$. Then, similarly to the previous case, in the colouring induced on sets of $r-1$ elements in our sequence, we can find a red $(k-1)$-clique or a blue $(\ell-1)$-clique. Adding x_{m+1} then gives us either a red k (r)-clique or a blue ℓ (r)-clique. This gives us the upper-bound

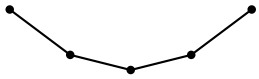
$$R_{(r)}(k,\ell) \leq 2^{\binom{m}{r-1}+1} \leq 2^{m^{r-1}} = 2^{R_{(r-1)}(k-1,\ell-1)^{r-1}}$$

(the $+1$ again takes care of the "small number"). □

6.1.5 Cups and Caps

We now move on to a result by Erdős and Szekeres.

Definition 6.1.11 (Cups and Caps). A k-**cap** and a k-**cup** are (respectively) concave and convex arrangements of k points.



a 5-cup



a 5-cap

We are interested in how many points it takes to force one of these.

Definition 6.1.12 (Erdős–Szekeres Number). The **Erdős–Szekeres number for k and ℓ**, denoted $\mathrm{ES}(k,\ell)$, is the smallest $n \in \mathbb{N}$ such that any n points in $\mathbb{R}^2$ in general position (with no two on a vertical line) have a k-cap or an ℓ-cup.

We will show, eventually, that

$$\mathrm{ES}(k,\ell) \leq 2\binom{k+\ell-4}{k-2} \leq 2^{k+\ell}$$

Lemma 6.1.13. *For all $k,\ell \geq 3$,*

$$\mathrm{ES}(k,\ell) \leq \mathrm{ES}(k-1,\ell) + \mathrm{ES}(k,\ell-1) - 1$$

Proof. We fix an arbitrary set $X \subseteq \mathbb{R}^2$ of $\mathrm{ES}(k-1,\ell)+\mathrm{ES}(k,\ell-1)-1$ points in general position, no two on a vertical line, and show that it contains a k-cap or an ℓ-cup.

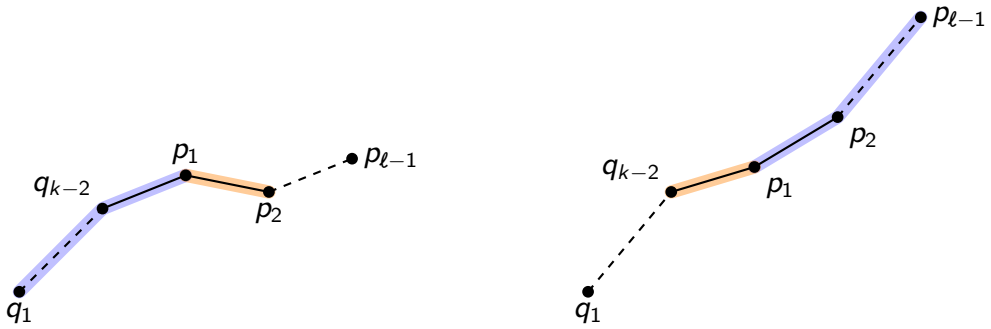
Deny. Let $L \subseteq X$ be the set of points in X that are the last points in some $(k-1)$-cap, so that $X \setminus L$ has no $(k-1)$-caps. Thus

$$|X \setminus L| < \mathrm{ES}(k-1,\ell)$$

Since $|X| = \mathrm{ES}(k-1,\ell) + \mathrm{ES}(k,\ell-1) - 1$, we conclude that $|L| \geq \mathrm{ES}(k,\ell-1)$.

As X has no k-cap, neither does L, so L must have an $(\ell-1)$-cup $p_1,\ldots,p_{\ell-1}$. Since $p_1 \in L$, it is the last point of some $(k-1)$-cap $q_1,\ldots,q_{k-1} = p_1$. One of these two extends, according to how the slopes compare at p_1: let s be the slope of $q_{k-2}p_1$ and t the slope of p_1p_2. These are

different, since otherwise q_{k-2}, p_1 and p_2 would lie on a line.



$s > t$: a k-cap $s < t$: an ℓ-cup

$\underline{s > t.}$ The slopes keep decreasing past p_1, so $q_1, \ldots, q_{k-1}, p_2$ is a k-cap.

$\underline{s < t.}$ The slopes are already increasing at p_1, so $q_{k-2}, p_1, p_2, \ldots, p_{\ell-1}$ is an ℓ-cup.

Either way, we have a contradiction. □

Corollary 6.1.14. *For all $k, \ell \geq 2$,*

$$\mathrm{ES}(k, \ell) \leq 2\binom{k+\ell-4}{k-2} \leq 2^{k+\ell}$$

Proof. We prove the first inequality by induction on $k + \ell$. When $k = 2$ or $\ell = 2$, any two points form both a 2-cap and a 2-cup, so $\mathrm{ES}(k, \ell) \leq 2 = 2\binom{k+\ell-4}{k-2}$.

Fix some $k, \ell \geq 3$ and assume the result holds for all k', ℓ' with $k' + \ell' < k + \ell$. We then get, by Lemma 6.1.13

$$\begin{aligned}\mathrm{ES}(k, \ell) &\leq \mathrm{ES}(k-1, \ell) + \mathrm{ES}(k, \ell-1) \\ &\leq 2\binom{k+\ell-5}{k-3} + 2\binom{k+\ell-5}{k-2} \\ &= 2\binom{k+\ell-4}{k-2}\end{aligned}$$

with the second inequality given by the induction hypothesis.

For the second inequality, $\binom{k+\ell-4}{k-2} \leq 2^{k+\ell-4}$, since a single binomial coefficient is at most the sum of its row of Pascal's triangle, so $2\binom{k+\ell-4}{k-2} \leq 2^{k+\ell-3} \leq 2^{k+\ell}$. □

6.2 Van der Waerden's Theorem

The rest of this chapter is motivated by the following question: if I r-colour the natural numbers $\mathbb{N}$, must I be able to find arbitrarily long monochromatic arithmetic progressions?

Definition 6.2.1 (van der Waerden's Number). For $k, r \in \mathbb{N}$, the **van der Waerden number for k and r**, denoted $\mathrm{VW}(k, r)$, is the least N such that any r-colouring of $[N]$ admits a k-term monochromatic arithmetic progression.

We will eventually show the following.

Theorem 6.2.2 (van der Waerden's Theorem). *For all $k, r \in \mathbb{N}$, $\mathrm{VW}(k, r)$ exists.*

$\mathrm{VW}(2, r) = r + 1$ is fairly straightforward. But what about $\mathrm{VW}(3, 2)$?

6.2.1 Colour-Focusing

We can show that $\mathrm{VW}(3, 2) \geq 9$, because of the colouring

1 2 3 4 5 6 7 8 9

where the minute you hit 9 you get a monochromatic AP: $1, 5, 9$ if 9 is red, and $7, 8, 9$ if 9 is blue. (The choices previously were to try and minimise the chance of getting a 3-term AP.)

We can also show that $\mathrm{VW}(3, 2) \leq 325$. How?

Write out blocks of five as follows:

1	2	3	4	5		1	2	3	4	5		1	2	3	4	5	⋯

(This is meant to represent a sequence of consecutive numbers, so the second box should really contain $6, 7, 8, 9, 10$, and so on. But this notation will prove convenient.)

We will essentially discover that when we have 65 such boxes, any 2-colouring of this sequence admits a monochromatic 3-term AP.

Fix a 2-colouring of this sequence. Notice that in any block of five, there is some 3-term AP such

that the first two terms are monochromatic. (Indeed, this is what motivates the choice of *five* in the first place.)

We have a total of 2^5 ways of colouring a block of five, so if we write down a total of $33 = 2^5 + 1$ blocks (which, in the worst case, consists of a block with each colour configuration, and then one more), we can always find two identically coloured blocks. Say these are the ith and the jth, with $i < j \leq 33$. Let $a, a+d, a+2d \in \{1,2,3,4,5\}$ be the positions of their 3-term AP, with a and $a+d$ red, say.[3] If $a+2d$ is also red, we're done, so suppose it is blue.

The intuition is to build an "AP of blocks" and use the next term after i and j to obtain the desired monochromatic 3-term AP. Indeed, if we start at block i, this AP looks like $i + \ell(j-i)$. Going to the $\ell = 2$ term should give us the $(2j-i)$th block. Let's now show that this works.

Write $D = 5(j-i)$ for the distance between the two blocks, so that the kth element of block j is exactly $D +$ (the kth element of block i). In the $(2j-i)$th block, the kth element is exactly D away from the kth element of block j. Now denote by x the $(a+2d)$th element of the $(2j-i)$th block.

x is red. Then position a of block i, position $a+d$ of block j and x form a red AP with difference $D+d$.

x is blue. Then position $a+2d$ of blocks i, j and $2j-i$ form a blue AP with difference D. (Remember, blocks i and j are coloured identically, and since the $(a+2d)$th element of block j is blue, so is that of i.)

Since $2j - i \leq 2 \cdot 33 - 1 = 65$, it suffices to have 65 blocks of five, ie, $5 \cdot 65 = 325$ numbers.

We will now apply this technique to find a general upper-bound for van der Waerden numbers, and thereby, show they always exist (ie, prove van der Waerden's theorem).

We want to emphasise the concept of *colour-focusing*: in our previous example, we saw that there were two APs that agreed at the last term, that were monochromatic until that one last "focus" term was coloured. That colouring would extend one of them but not the other. We thus make

[3] Hence the notational convenience of using the same numbers, $1, 2, 3, 4, 5$, for each block.

the following definition.

Definition 6.2.3 (Colour-Focused APs)**.** A collection of k-term APs is **colour-focused** if the following both hold:

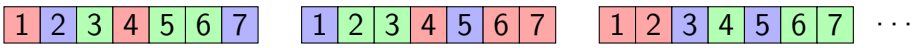
- each AP is monochromatic
- all the APs have a common $(k+1)$th term

The common $(k+1)$th term of a family of colour-focused APs is called their **focus point**.

The focus point could be the same colour as one of the APs, but it does not have to be. For our purposes, we will be interested in colour-focused APs that are different colours.

Let's now try and repeat the $\mathrm{VW}(3,2)$ argument to find an upper-bound for $\mathrm{VW}(3,3)$.

Divide the natural numbers into blocks of 7. Again, our choice is motivated by the pigeonhole principle: for any 3-colouring of all the numbers, any 7-block admits a 3-term AP such that the first 2 terms are monochromatic.



We have a total of 3^7 ways of colouring a block, so if we write down $3^7+1 = 2188$ blocks, there are bound to be two that are identically coloured. As before, if we double this number and subtract one, ie, if we have $2(3^7+1)-1 = 4375$ blocks, then we are guaranteed to have two colour-focused APs with a focus point.

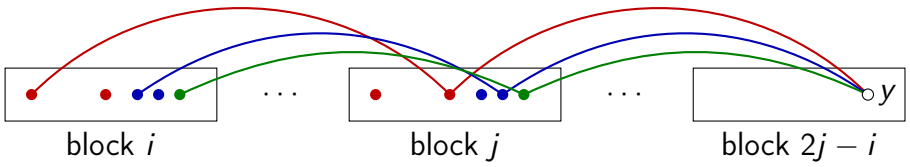
The problem is that this focus point could take *three* values, but as we only have two monochromatic APs, if it takes the colour not taken by the two APs, we do not have a monochromatic 3-term AP.

What we have shown above is that if we have a total of 4375 blocks, that is, $7 \cdot 4375$ numbers, we are guaranteed to have two APs with a shared focus point, which may or may not complete them.

To get three colour-focused APs, we need to "go up one level" and treat blocks of $7 \cdot 4375 = 30625$ numbers the way we treated blocks of 7 to get two colour-focused APs. By the above, each such block contains either a monochromatic 3-term AP, in which case we're done, or two colour-focused APs of different colours, say red and blue, whose focus point is green. There are 3^{30625} ways of

colouring a 30625-block, so among $3^{30625}+1$ of them, two are identically coloured, say the ith and the jth. Write $\Delta = 30625(j-i)$, and let y be the element of the $(2j-i)$th block in the same position as the focus point. We now have three APs heading for y:

- the red AP, with its first term taken from block i and its second from block j;
- the blue AP, likewise;
- the focus point of block i followed by the focus point of block j, both green.



Each of these is monochromatic, they have different colours, and each continues to y: if p, q is an AP in block i whose next term is the focus point f, then p and $q+\Delta$ form an AP whose next term is $f+2\Delta = y$. So these are three colour-focused APs of different colours, and whatever the colour of y, it completes one of them to a monochromatic 3-term AP. As before, it suffices to have $2\,(3^{30625}+1)-1$ blocks of 30625 numbers, so

$$\mathrm{VW}(3,3) \leq 30625 \cdot \left(2 \cdot 3^{30625}+1\right)$$

Not only have we found a bound on $\mathrm{VW}(3,2)$ and $\mathrm{VW}(3,3)$, we've also shown they *exist*. We can actually use the strategy above to prove van der Waerden's Theorem (Theorem 6.2.2).

6.2.2 Colour-Focus Numbers

First, a general definition.

Definition 6.2.4 (Colour-Focus Number)**.** For k, r, s with $s \leq r$, define the **colour-focus number** $\mathrm{CF}(k, r, s)$ to be the least N such that any r-colouring of $[N]$ has either a monochromatic $(k+1)$-term AP or s colour-focused k-term APs of different colours, with focus point in $[N]$.

We can bound van der Waerden numbers above by colour-focus numbers as follows.

Lemma 6.2.5. *For all $k, r \in \mathbb{N}$,*

$$\mathrm{VW}(k, r) \leq \mathrm{CF}(k-1, r, r)$$

Proof. Let $N = \mathrm{CF}(k-1, r, r)$. We need to show that any r-colouring of $[N]$ contains a k-term monochromatic arithmetic progression. But by definition of the colour-focus number, either $[N]$ admits a k-term monochromatic arithmetic progression (in which case we're done) or r colour-focused $(k-1)$-term arithmetic progressions. As we argued above, if there are r such APs, then the focus must complete one of them, as we are r-colouring. So again we're done. □

We now prove a recursive bound on the colour-focus numbers.

Lemma 6.2.6. *For all $k, r, s \in \mathbb{N}$ with $2 \leq s \leq r$,*

$$\mathrm{CF}(k, r, s) \leq \mathrm{CF}(k, r, s-1) \cdot 2\,\mathrm{VW}\left(k, r^{\mathrm{CF}(k,r,s-1)}\right)$$

Proof. Fix an $N \in \mathbb{N}$ and an r-colouring of $[N]$. Let $C = \mathrm{CF}(k, r, s-1)$. Consider blocks in $[N]$ of length C. If any of them contains a $(k+1)$-term monochromatic arithmetic progression, we're done, so assume that they all contain $(s-1)$ k-term colour-focused arithmetic progressions of different colours. As they are all of length C, they must all have one property or the other.

Each block can be coloured in one of r^C ways. If we took $\mathrm{VW}\left(k, r^C\right)$ many blocks, say $B_1, \ldots, B_{\mathrm{VW}\left(k, r^C\right)}$, by definition of VW, we would be able to find a k-term "arithmetic progression of blocks"

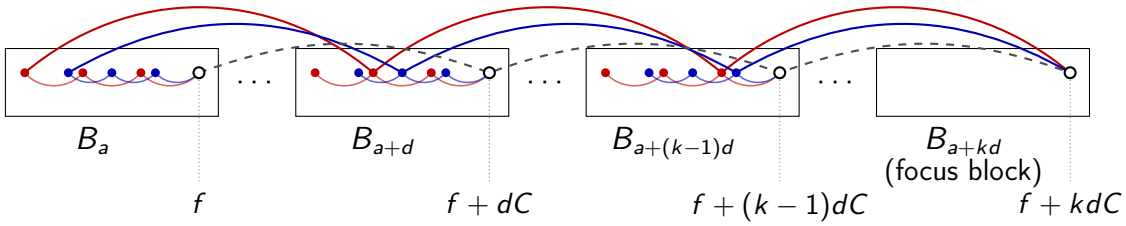
$$B_a, B_{a+d}, B_{a+2d}, \ldots, B_{a+(k-1)d}$$

that are all identically coloured.[4] If we take $2\,\mathrm{VW}\left(k, r^C\right)$ blocks[5], we also have the focus block B_{a+kd}.

[4] We can "code APs of blocks as APs", and in this coding, a sequence of numbers being monochromatic translates to blocks being identically coloured.

[5] For $k \geq 2$, $2\,\mathrm{VW}\left(k, r^{\mathrm{CF}(k,r,s-1)}\right) - 1$ would suffice, and we can even get tighter bounds, but the expression looks nicer without the -1.

What are some properties of this focus block? We know that it may not be coloured identically to the others in the AP of blocks. However, every block (APs and focus alike) has exactly C elements, so each block contains $(s-1)$ k-term colour-focused APs. Then, the families of APs that looks like "the jth term of the jth block", I get an AP with a focus point in B_{a+kd}. Ie, suppose my $(s-1)$ k-term APs are denoted $\left(x_j^i\right)_{1\le j\le k}$ for $1 \le i \le s-1$, then I can build new APs $\left(y_j^i\right)_{1\le j\le k}$ where for each i in my family, I define y_j^i to be the instance of x_j^i living in the jth block $B_{a+(j-1)d}$. This gives me $(s-1)$ colour-focused APs in my set of numbers from the $2\,\mathrm{VW}\left(k, r^C\right)$ blocks (the focus point being the element of the focus block occupying the position of the focus points in the other blocks). Moreover, when I take the AP consisting of all the focus points, I get yet another AP with the same focus point as the $\left(y_j^i\right)$. Its colour is different from theirs: the $\left(y_j^i\right)$ have the colours of the $\left(x_j^i\right)$, and if the focus point of B_a had the same colour as one of the $\left(x_j^i\right)$, the two together would form a monochromatic $(k+1)$-term AP, which we have ruled out. Thus, I get a total of s colour-focused APs.



The total number of elements living in my $2\,\mathrm{VW}\left(k, r^C\right)$ blocks is $C \cdot 2\,\mathrm{VW}\left(k, r^C\right)$. So effectively, when $N = C \cdot 2\,\mathrm{VW}\left(k, r^C\right)$, any r-colouring of $[N]$ admits either a $(k+1)$-term monochromatic AP or s colour-focused k-term APs of different colours. Thus, this value of N is an upper-bound on $\mathrm{CF}(k, r, s)$. □

We can use our recursive bound on colour-focus numbers to prove that colour-focus numbers always exist[6]. This ultimately tells us that van der Waerden numbers also always exist.

Theorem 6.2.7. $\mathrm{CF}(k, r, s)$ *always exists.*

Proof. Proceed by induction on k, keeping r and s generic[7]. When $k = 1$, the result is trivial,

[6] Maybe you need to choose k, r, s so that $\mathrm{CF}(k, r, s)$ actually makes sense, but we won't split hairs about this.
[7] In Lean this would be `induction k generalizing r s`.

because no matter the value of r or s, if we consider the set $[s+1]$, we can always find either a 2-term monochromatic AP (ie, two numbers with the same colour) or s 1-term monochromatic colour-focused APs (ie, $s+1$ numbers with distinct colours).

Now fix k and assume $\mathrm{CF}(k, r, s)$ exists for all r, s. We want to show that $\mathrm{CF}(k+1, r, s)$ also exists for all r, s. We proceed by induction on s.

$\mathrm{CF}(k+1, r, 1)$ certainly always exists. In general, $\mathrm{CF}(a, b, 1)$ exists iff there is some $N \in \mathbb{N}$ such that any b-colouring of $[N]$ admits a monochromatic a-term AP. Take $N \geq \mathrm{CF}(k, r, r)$ (which exists by the induction hypothesis on k). Then either $[N]$ admits a $(k+1)$-term monochromatic AP (in which case we're done) or $[N]$ admits r colour-focused k-term APs (in which case the focus point extends one of them to a monochromatic $(k+1)$-term AP, as there are precisely r of them).

Now fix s and suppose $\mathrm{CF}(k+1, r, s)$ exists for all r. Then by Lemmas 6.2.5 and 6.2.6, we know

$$\begin{aligned}\mathrm{CF}(k+1, r, s+1) &\leq \mathrm{CF}(k+1, r, s) \cdot 2\,\mathrm{VW}\left(k+1, r^{\mathrm{CF}(k+1,r,s)}\right) \\ &\leq \mathrm{CF}(k+1, r, s) \cdot 2\,\mathrm{CF}\left(k, r^{\mathrm{CF}(k+1,r,s)}, r^{\mathrm{CF}(k+1,r,s)}\right)\end{aligned}$$

The above colour-focus numbers both exist by the induction hypotheses on k and s, finishing the induction. □

Theorem 6.2.2 (van der Waerden's theorem) is an immediate corollary.

6.2.3 A Lower Bound from the Local Lemma

The bounds above are enormous. For comparison, here is what the Lovász local lemma (Corollary 5.1.3) gives from below.

Proposition 6.2.8. *For all $k \geq 2$,*

$$\mathrm{VW}(k, 2) > \frac{(k-1)\,2^{k-1}}{ek^2}$$

In particular, $\mathrm{VW}(k, 2) = \Omega\left(\frac{2^k}{k}\right)$.

Proof. Let $N = \left\lfloor \frac{(k-1)2^{k-1}}{ek^2} \right\rfloor + 1$. It suffices to find a 2-colouring of $[N]$ with no monochromatic

k-term arithmetic progression, as then $\mathrm{VW}(k,2) > N$. If $N < k$ there is nothing to find, since $[N]$ has no k-term progressions at all, so assume $N \geq k$.

Colour $[N]$ red or blue uniformly and independently at random. For each k-term arithmetic progression $P \subseteq [N]$, let A_P be the event that P is monochromatic, so that $\Pr(A_P) = 2^{1-k}$. A_P is determined by the colours of the points of P alone, so joining A_P to A_Q exactly when P and Q share a point gives a dependency graph. A progression through a given point is determined by its common difference, which is at most $\frac{N-1}{k-1}$, and by the position of the point in it, so a point lies in at most $\frac{k(N-1)}{k-1}$ progressions, and A_P has at most

$$D = \frac{k^2(N-1)}{k-1} - 1$$

neighbours. Now

$$e \cdot 2^{1-k}(D+1) = \frac{ek^2(N-1)}{(k-1)2^{k-1}} \leq 1$$

by the choice of N, so Corollary 5.1.3 applies, and with positive probability no A_P occurs: some 2-colouring of $[N]$ has no monochromatic k-term arithmetic progression. □

6.3 The Hales–Jewett Theorem

Van der Waerden's theorem is about progressions in the integers. The Hales–Jewett theorem strips the arithmetic away and keeps only the combinatorics of the colour-focusing argument, and it is the stronger result: at the end of this section it will give van der Waerden's theorem back, and other things with it.

6.3.1 Combinatorial Lines

Combinatorial lines are what Hales–Jewett replaces arithmetic progressions with, and the hypercube $[n]^d$—the set of strings of length d over the alphabet $[n]$—is where they live.

Definition 6.3.1 (Combinatorial Line). Fix $n, d \in \mathbb{N}$ and a symbol $x \notin [n]$. A **template** is a string $\sigma \in ([n] \cup \{x\})^d \setminus [n]^d$, ie, one in which x occurs at least once; the coordinates at which σ is x are its **active coordinates**, and the others are **fixed**. For $i \in [n]$, write

$\sigma(i) \in [n]^d$ for the string obtained from σ by replacing every x by i. The **combinatorial line** with template σ is

$$L_\sigma = \{\sigma(1), \sigma(2), \ldots, \sigma(n)\}$$

and $\sigma(n)$ is its **last point**. We will usually just say **line**.

In $[3]^2$, for instance, the templates $x1$, $2x$ and xx give two lines parallel to the axes and the main diagonal, whereas the anti-diagonal $\{13, 22, 31\}$ is a line in the geometric sense but not a combinatorial one, since its two coordinates move in opposite directions. The number we are after is the dimension that forces a monochromatic line.

Definition 6.3.2 (Hales-Jewett Number)**.** For all $n, r \in \mathbb{N}$, define $\mathrm{HJ}(n, r)$ to be the smallest number d such that any r-colouring of the hypercube $[n]^d$ has a monochromatic line.

As with van der Waerden's numbers, we will eventually show that these always exist.

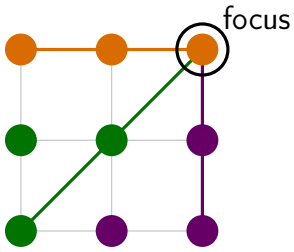
Theorem 6.3.3 (Hales–Jewett Theorem)**.** *For all $n, r \in \mathbb{N}$, $\mathrm{HJ}(n, r)$ exists.*

6.3.2 Colour-Focused Lines

Colour-focusing works for lines exactly as it did for arithmetic progressions.

Definition 6.3.4 (Colour-Focused Lines)**.** Given combinatorial lines $L_1, L_2, \ldots, L_s$, they are **colour-focused** if their last points coincide and each is monochromatic (in a distinct colour) on all but the last point.

For example, draw $[3]^2$ as a grid, with the first coordinate running left to right and the second bottom to top. In the colouring below, the top row, the right-hand column and the diagonal are three colour-focused lines: each is monochromatic on its first two points, in orange, purple and green respectively, and all three end at the top-right point, which is their focus. Whatever colour that point gets, it completes one of them; here it is orange, and the top row is a monochromatic line.



Definition 6.3.5 (Focused Hales-Jewett Number). FHJ(n, r, s) is the minimum d such that any r-colouring of $[n]^d$ has s colour-focused lines or a monochromatic line.

We can see that

Lemma 6.3.6. *For all $n, r \in \mathbb{N}$,*

$$\mathrm{HJ}(n, r) \leq \mathrm{FHJ}(n, r, r)$$

Proof. Let $d = \mathrm{FHJ}(n, r, r)$ and fix an r-colouring of $[n]^d$. If it has a monochromatic line we are done. Otherwise it has r colour-focused lines, of r distinct colours, with a common last point. That point has one of the r colours, so it completes the line of that colour to a monochromatic one. □

We will also show

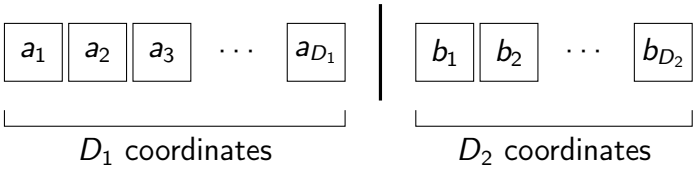
Lemma 6.3.7. *For all $n, r, s \in \mathbb{N}$ with $2 \leq s \leq r$,*

$$\mathrm{FHJ}(n, r, s) \leq \mathrm{FHJ}(n, r, s-1) + \mathrm{HJ}\left(n-1, r^{n^{\mathrm{FHJ}(n,r,s-1)}}\right)$$

Proof. For this argument, denote $\mathrm{FHJ}(n, r, s-1)$ by D_2 and $\mathrm{HJ}\left(n-1, r^{n^{\mathrm{FHJ}(n,r,s-1)}}\right)$ by D_1. (The letters are apt because these are dimensions.) We show essentially that $\mathrm{FHJ}(n, r, s) \leq D_2 + D_1$ and note that $D_1 = \mathrm{HJ}\left(n-1, r^{n^{D_2}}\right)$.

The inequality is essentially saying "any r-colouring of the $[n]^{D_1+D_2}$ cube contains s colour-focused lines or a monochromatic line."

A point in my cube has coordinates



I'm given a colouring of the $[n]^{D_1+D_2}$ cube. Any string $a_1, a_2, \ldots, a_{D_1}$ completes to n^{D_2} points in $[n]^{D_1+D_2}$. Ie, there are n^{D_2} ways of completing it, ie, there are n^{D_2} points in $[n]^{D_1+D_2}$ whose first D_1 coordinates are these. So, taken together, the n^{D_2} completions of a given string can be coloured in one of $r^{n^{D_2}}$ ways, and which of these ways our colouring picks depends on the string.

In fact, we observe that *any r-colouring of $[n]^{D_1+D_2}$ involves an $r^{n^{D_2}}$-colouring of the $[n]^{D_1}$ cube.*

Since $D_1 = \mathrm{HJ}(n-1, r^{n^{D_2}})$, there is a combinatorial line $\mathcal{L}_1 \subseteq [n]^{D_1}$ whose first $n-1$ points all get the same colour from all of the $r^{n^{D_2}}$ choices.

$\mathcal{L}_1$ corresponds to a string of the form

$$\sigma \in ([n-1] \cup \{x\})^{D_1} \setminus [n]^{D_1}$$

(for some symbol x), such that the $r^{n^{D_2}}$-colouring agrees on $\sigma(1), \ldots, \sigma(n-1)$. Unwinding what that colouring is, this says that for all $i, i' \in [n-1]$ and all $b \in [n]^{D_2}$, the points $(\sigma(i), b)$ and $(\sigma(i'), b)$ of our cube have the same colour. Write χ for the given r-colouring of $[n]^{D_1+D_2}$, and $\chi'(b) = \chi(\sigma(1), b)$ for that common colour. Then χ' is an r-colouring of $[n]^{D_2}$, and $D_2 = \mathrm{FHJ}(n, r, s-1)$, so $[n]^{D_2}$ contains a χ'-monochromatic line or $s-1$ χ'-colour-focused lines.

$[n]^{D_2}$ has a χ'-monochromatic line L_τ. Consider the template $(\sigma(1), \tau)$, whose active coordinates are those of τ and whose first D_1 coordinates are fixed at $\sigma(1)$. Its line has points $(\sigma(1), \tau(i))$ for $i \in [n]$, of colour $\chi(\sigma(1), \tau(i)) = \chi'(\tau(i))$, which does not depend on i. So $[n]^{D_1+D_2}$ has a monochromatic line and we are done.

$[n]^{D_2}$ has $s-1$ χ'-colour-focused lines $L_{\tau_1}, \ldots, L_{\tau_{s-1}}$. Let $f \in [n]^{D_2}$ be their common last point and $c_1, \ldots, c_{s-1}$ their colours, which are distinct. If $\chi'(f) = c_j$ for some j, then L_{τ_j} is χ'-monochromatic and the previous case applies, so suppose not. For each j, the template (σ, τ_j) has as its active coordinates those of σ together with those of τ_j, and its line has points $(\sigma(i), \tau_j(i))$ for $i \in [n]$. For $i \leq n-1$, the colour of such a point is $\chi(\sigma(i), \tau_j(i)) =$

$\chi'(\tau_j(i)) = c_j$, since $\tau_j(i)$ is one of the first $n-1$ points of L_{τ_j}; and its last point is $(\sigma(n), f)$. Alongside these, the template (σ, f), with the last D_2 coordinates fixed at f, gives a line with points $(\sigma(i), f)$, of colour $\chi(\sigma(i), f) = \chi'(f)$ for $i \leq n-1$, and with the same last point $(\sigma(n), f)$. These s lines are monochromatic on all but their last point, in the s distinct colours $c_1, \ldots, c_{s-1}, \chi'(f)$, and their last points coincide: they are s colour-focused lines in $[n]^{D_1+D_2}$.

Either way, $[n]^{D_1+D_2}$ has a monochromatic line or s colour-focused lines, so $\mathrm{FHJ}(n, r, s) \leq D_1 + D_2$. □

With the recursion in hand, the Hales–Jewett theorem goes exactly as van der Waerden's did: induct on the size of the alphabet, and inside that on the number of focused lines.

Proof of Theorem 6.3.3. Induct on n, keeping r generic. When $n = 1$, the cube $[1]^1$ is a single point, which is a line (its template is x) and is monochromatic under any colouring, so $\mathrm{HJ}(1, r) = 1$.

Now fix $n \geq 2$ and assume that $\mathrm{HJ}(n-1, r')$ exists for every $r' \in \mathbb{N}$. By Lemma 6.3.6 it suffices to show that $\mathrm{FHJ}(n, r, s)$ exists for all r and all $1 \leq s \leq r$, which we do by induction on s. For $s = 1$, take $d = \mathrm{HJ}(n-1, r)$: any r-colouring of $[n]^d$ restricts to one of $[n-1]^d$, which has a monochromatic line with some template $\sigma \in ([n-1] \cup \{x\})^d$, and read in $[n]^d$ the same template gives a line whose first $n-1$ points are that line. It is therefore monochromatic on all but its last point, which is to say that it is one colour-focused line, and so $\mathrm{FHJ}(n, r, 1) \leq \mathrm{HJ}(n-1, r)$. For $2 \leq s \leq r$, Lemma 6.3.7 bounds $\mathrm{FHJ}(n, r, s)$ by $\mathrm{FHJ}(n, r, s-1)$, which exists by the induction on s, plus $\mathrm{HJ}\left(n-1, r^{n^{\mathrm{FHJ}(n,r,s-1)}}\right)$, which exists by the induction on n. □

6.3.3 From Lines to Progressions

The theorem gives us a monochromatic copy of any finite pattern, in any number of colours and any dimension.

Theorem 6.3.8. *For any finite $X \subseteq \mathbb{Z}^k$ and any $r \in \mathbb{N}$, any r-colouring of $\mathbb{Z}^k$ contains a copy $\alpha X + \beta$ of X, for some $\alpha \in \mathbb{N}$ and $\beta \in \mathbb{Z}^k$, that is monochromatic.*

Proof. Let $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Z}^k$, so that $n = |X|$, let $d = \mathrm{HJ}(n, r)$, which exists by Theorem 6.3.3, and fix an r-colouring χ of $\mathbb{Z}^k$. Define

$$f : [n]^d \to \mathbb{Z}^k : (a_1, \ldots, a_d) \mapsto x_{a_1} + x_{a_2} + \cdots + x_{a_d}$$

and give each point $a \in [n]^d$ the colour $\chi(f(a))$ of its image. This is an r-colouring of $[n]^d$, so by the choice of d it has a monochromatic line L_σ. Let S be the set of active coordinates of σ and $\alpha = |S|$, so that $\alpha \geq 1$. For each $i \in [n]$, the string $\sigma(i)$ has i in the α active coordinates and the entries of σ itself in the rest, so

$$f(\sigma(i)) = \sum_{j \notin S} x_{\sigma_j} + \alpha x_i = \beta + \alpha x_i$$

where $\beta = \sum_{j \notin S} x_{\sigma_j} \in \mathbb{Z}^k$ does not depend on i. Thus $f(L_\sigma) = \{\beta + \alpha x_i \mid i \in [n]\} = \alpha X + \beta$, and since every point of L_σ has the same colour, so does every point of $\alpha X + \beta$ under χ. □

This is known as Gallai's theorem, and [2, Chapter 2] deduces it from the Hales–Jewett theorem of [3] in exactly this way.

To find bounds on van der Waerden numbers in terms of Hales-Jewett numbers, let's ask ourselves the following question: can we map from $[n]^d$ to some line in a way that takes geometric/combinatorial lines to arithmetic progressions?

One thing we immediately see is that if we consider the mapping

$$\phi : [n]^d \to \left[n^d\right] : p \mapsto 1 + (p_1 - 1) + (p_2 - 1)\, n + \cdots + (p_d - 1)\, n^{d-1}$$

which reads a point as the digits of a number written in base n, then lines go to progressions: if S is the set of active coordinates of a template σ, then

$$\phi(\sigma(i)) = 1 + \sum_{j \notin S} (\sigma_j - 1)\, n^{j-1} + (i - 1) \sum_{j \in S} n^{j-1}$$

so as i runs over $[n]$ the images $\phi(\sigma(1)), \ldots, \phi(\sigma(n))$ form an n-term arithmetic progression with common difference $\sum_{j \in S} n^{j-1} > 0$. An r-colouring of $\left[n^d\right]$ therefore pulls back along ϕ to an r-colouring of $[n]^d$, and when $d = \mathrm{HJ}(n, r)$ the monochromatic line this contains maps to a

monochromatic n-term progression. In other words,

$$\mathrm{VW}(n, r) \leq n^{\mathrm{HJ}(n,r)}$$

so that the Hales–Jewett theorem gives van der Waerden's theorem a second time.

Nothing about this used the particular form of ϕ beyond one property: a line is sent to a progression that does not degenerate, because its active coordinates all move together, so the image of a line can never collapse to a point. Any map with that property gives a Ramsey theorem for whatever a line maps to.

Similarly, if we do

$$\gamma : [n]^d \to \left[(n-1)^2 d + 1\right] : p \mapsto 1 + (p_1 - 1)^2 + (p_2 - 1)^2 + \cdots + (p_d - 1)^2$$

the image of a line under γ is now a "quadratic progression".

6.4 Infinite Ramsey Theory

Consider the complete graph on the natural numbers, denoted $K_{\mathbb{N}}$. First, observe that any 2-colouring of the edges of $K_{\mathbb{N}}$ has an infinite monochromatic clique: indeed, pick $v_1 \in \mathbb{N}$; infinitely many of the edges at v_1 have the same colour, say c_1, so let S_1 be the infinite set of their other endpoints. Pick $v_2 \in S_1$; infinitely many of the edges from v_2 into S_1 have the same colour, say c_2, and let $S_2 \subseteq S_1$ be the set of their other endpoints. Continuing in this way gives $v_1, v_2, \ldots$ and colours $c_1, c_2, \ldots$ such that the edge $v_i v_j$ has colour c_i whenever $i < j$. One colour occurs infinitely often among the c_i, and the v_i with that c_i form an infinite monochromatic clique.

Ramsey numbers of finite sets have a strengthening in which the monochromatic set is asked to be large relative to its own elements.

Definition 6.4.1 (Large Ramsey Number)**.** A finite set $X \subseteq \mathbb{N}$ is **relatively large** if $|X| \geq \min X$. For $n, k, c \in \mathbb{N}$, define $\bar{R}_k^{(c)}(n)$ to be the smallest $N \in \mathbb{N}$ such that any c-colouring of the k-subsets of $[N]$ has a relatively large set $X \subseteq [N]$ with $|X| \geq n$ all of whose k-subsets have the same colour.

So X is a monochromatic clique in the sense of hypergraph Ramsey numbers, and for $c = 2$ the definition without the words "relatively large" would be $R_{(k)}(n, n)$. The extra condition is what makes the number interesting: a monochromatic clique whose least element is m now has to have at least m elements, so the further along it starts, the larger it has to be. That these numbers exist at all follows from the infinite version of Ramsey's theorem by a compactness argument, which we will not give.

One interesting theorem (that we will not prove here) is that the existence of the above cannot be proven in Peano arithmetic. This is the Paris–Harrington theorem [4], the first example of a natural combinatorial statement that is true but not provable from Peano's axioms.

Chapter 7

Discrete Geometry

We now put points in the plane and ask how many pairs of them can be at distance exactly 1. In the language of Definition 2.3.1, this asks how many edges a subgraph of the unit distance graph on n vertices can have, rather than how many colours it needs. Constructions give lower bounds; the upper bound will come from drawing graphs in the plane and counting how often their edges are forced to cross. The same counting then bounds how often points can lie on lines. The rest of the chapter is about lines in $\mathbb{R}^3$ and over finite fields, and its tool is polynomials of low degree.

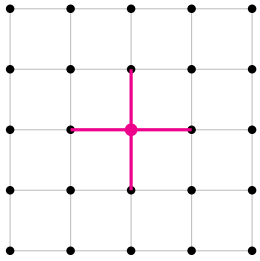
7.1 The Erdős Unit Distance Problem

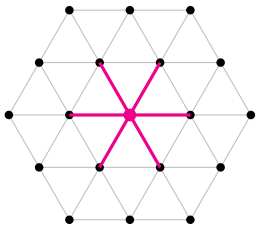
The Erdős Unit Distance Problem asks the following question:

How many unit distances can there be amongst n distinct points in $\mathbb{R}^2$?

7.1.1 Lower Bounds

Some naïve constructions include making each point adjacent to four or six others by way of square/triangular tiling.





Another construction gives us even better numbers.

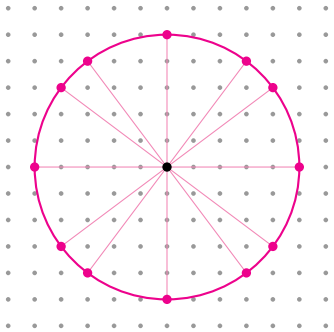
Take a square. Each vertex is connected to 2 others. Translate it by one unit and connect it. Then each vertex is connected to 3 others. Translate it by another unit and connect it again. Then each vertex is connected to 4 others. Keep going. In this construction, you get $n\log(n)$ as your lower-bound.

More precisely, each translation is by a unit vector in a new direction, and the directions are chosen generically enough (a random choice works) that no two points ever coincide. The square is a 2-dimensional cube, and each translation adds a dimension, so after $d-2$ translations we have $n=2^d$ points, each at unit distance from d others. Each unit distance is counted from both of its ends, so there are at least

$$\frac{nd}{2}=\frac{1}{2}n\log_2(n)$$

of them.

Erdős gave something even better: draw a grid with the points some distance $\frac{1}{R}$ apart, where R has the property that $R^2=a^2+b^2$ in many ways. The point is that this looks like the following:



Here $R=5$, and $25=0^2+5^2=3^2+4^2=4^2+3^2$ puts 12 grid points on the circle of radius R grid units, ie, at unit distance from the centre.

We can then do some very nice algebraic number theory involving $\mathbb{Z}[i]$ to get

$$n^{1+\frac{c}{\log(\log(n))}}$$

Erdős conjectured that the maximum you could get to is $n^{1+o(1)}$. OpenAI disproved this by finding an $n^{1+\delta}$ construction for some $\delta > 0$.

We will not get into the lower-bounds here, but we will show an upper-bound: $O\left(n^{\frac{4}{3}}\right)$. The tool for it is drawing graphs in the plane.

7.1.2 Planar Graphs

We also make a distinction between plane and planar graphs: planar graphs are abstract graphs with drawings in the plane, and plane graphs are the actual drawings themselves. More formally, we define the following.

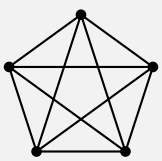
Definition 7.1.1 (Planar Graph). A graph G is **planar** if it has an **embedding** in the plane: an injective map sending each vertex to a point of $\mathbb{R}^2$ and each edge $\{u, v\}$ to a curve joining the images of u and v, such that no curve passes through the image of any other vertex and no two curves meet except at a common endpoint.

The picture itself is then the following.

Definition 7.1.2 (Plane Graph). A **plane graph** is the image of an embedding of a planar graph: the points and curves in $\mathbb{R}^2$ themselves. Its **faces** are the connected components of what remains of $\mathbb{R}^2$ once the plane graph is removed.

Not all graphs are planar.

Non-Example 7.1.3 (Non-planar graphs). K_5, K_{100}, any K_n for $n \geq 5$.



The reason graphs fail to be planar is that too many of their edges cross.

We follow the lecture's letters for the sizes of a graph.

Local Notation. For the rest of this section, a graph G has n vertices and m edges, and a plane graph has f faces. Where several graphs are in play, we write $n(H)$ and $m(H)$ for the numbers of vertices and edges of H.

(Connected) plane graphs satisfy Euler's Formula: $n - m + f = 2$. We can prove this using induction on m and some handwaving (one should say "visual intuition"), but to actually do this rigorously takes a lot of machinery (eg. the Jordan Curve Theorem).

Theorem 7.1.4 (Euler's Formula). *For a connected plane graph with n vertices, m edges and f faces,*

$$n - m + f = 2$$

Proof. Induction on m. `sorry` □

Notice that in any simple plane graph with at least 3 edges, we *must* have

- Every face is incident with ≥ 3 edges
- Every edge is incident with ≤ 2 faces

This gives us an important inequality.

Proposition 7.1.5. *For any connected simple plane graph with n vertices, m edges and f faces, if $m \geq 3$, then*

$$m \leq 3n - 6$$

Proof. Count edge-face incidences. By above,

$$3f \leq \sum_{F \text{ a face}} \#\{\text{edges } F \text{ is incident with}\} = \sum_{E \text{ an edge}} \#\{\text{faces } E \text{ is incident with}\} \leq 2m$$

Thus, we get the equivalent inequalities

$$3f \leq 2m \qquad \Longleftrightarrow \qquad f \leq \frac{2m}{3}$$

Combining with Euler's formula, we get

$$2 = n - m + f \leq n - m + \frac{2m}{3} = n - \frac{m}{3}$$

Rearranging gives us the desired inequality. □

7.1.3 Crossing Numbers

A graph that is not planar can still be drawn in the plane, and we can ask how few crossings it needs.

Definition 7.1.6 (Crossing Number)**.** For every finite simple graph G, define the **crossing number**, denoted $\mathrm{cr}(G)$, to be the minimum number of crossings between edges, taken over all drawings of G in the plane. Ie, define

$$\mathrm{cr}(G) = \min_{\text{drawings of } G \text{ in the plane}} \#\{\text{crossings between edges}\}$$

We might expect $\mathrm{cr}(K_n) = O\Big((\#\text{edges})^2\Big) = O(n^4)$.

Proposition 7.1.5 gives us a first lower bound.

Lemma 7.1.7. *If G is simple, then $\mathrm{cr}(G) \geq m - 3n$.*

Proof. First, every simple plane graph H has $m(H) \leq 3n(H)$. Indeed, each connected component of H is a connected simple plane graph in its own right: a component with n_i vertices and $m_i \geq 3$ edges has $m_i \leq 3n_i - 6 \leq 3n_i$ by Proposition 7.1.5, and one with $m_i \leq 2$ edges has $m_i \leq 3n_i$ since $n_i \geq 1$. Summing over the components gives $m(H) \leq 3n(H)$.

Now take a drawing of G with exactly $\mathrm{cr}(G)$ crossings, and for each crossing delete one of the two edges involved. We have deleted at most $\mathrm{cr}(G)$ edges, and what remains is a drawing with no crossings at all, ie, a simple plane graph on the same n vertices with at least $m - \mathrm{cr}(G)$ edges.

Thus $m - \text{cr}(G) \leq 3n$, which rearranges to $\text{cr}(G) \geq m - 3n$. □

Consider the case $G = K_n$. So $m = \binom{n}{2}$. The above inequality gives us

$$\text{cr}(K_n) \geq \binom{n}{2} - 3n$$

But we can probably do better than this: we'd at least expect each edge to cross each other edge or something, so we might think $\text{cr}(K_n) = \Omega(n^4)$.

7.1.4 The Crossing Lemma

Probabilistic methods to the rescue!

Local Notation. Given a graph G and $p \in [0, 1]$, define G_p to be the graph that retains vertices of G independently with probability p (ie, that deletes vertices of G independently with probability $1 - p$). Define random variables n_p and m_p, respectively denoting the number of vertices and edges in G_p. Denote by n and m the number of vertices and edges respectively of G.

We actually want to think of G_p as the space of all subgraphs of G, with the power sigma algebra, with each singleton (ie, the probability of each subgraph being realised) having probability determined by p.

Lemma 7.1.8. *$\mathbb{E}(n_p) = p \cdot n$ and $\mathbb{E}(m_p) = p^2 \cdot m$.*

Proof sketch. $\mathbb{E}(n_p) = p \cdot n$ is clear. To compute $\mathbb{E}(m_p)$, define indicator random variables ξ_e for each edge e, so that m_p is the sum over all edges e of the ξ_e. Then apply linearity of expectation. □

We can relate $\text{cr}(G_p)$ with $\text{cr}(G)$.

Lemma 7.1.9. *$\mathbb{E}(\text{cr}(G_p)) \leq p^4 \cdot \text{cr}(G)$*

Proof. `sorry` □

If we can then find a tighter lower-bound for $\mathrm{cr}(G_p)$, we can get a tighter lower-bound for $\mathrm{cr}(G)$. If we then optimise it by finding the p that makes it tightest, we can actually do a bit better than what we've done so far.

This is the crossing lemma.

Theorem 7.1.10 (The Crossing Lemma). *If G is a simple graph with n vertices and $m \geq 4n$ edges, then*

$$\mathrm{cr}(G) \geq \frac{m^3}{64n^2}$$

Proof. `sorry` □

7.1.5 A Unit-Distance Upper-Bound

We can now give the upper-bound we promised.

Theorem 7.1.11. *Among any n points in $\mathbb{R}^2$, there are $O\left(n^{\frac{4}{3}}\right)$ unit distances.*

Proof. `sorry` □

7.2 Point–Line Incidences

The crossing lemma also bounds how often points of the plane can lie on lines, and the bound in turn says something about sums and products of integers.

7.2.1 The Szemerédi–Trotter Theorem

Given finite sets $\mathcal{L}$ of lines and P of points in $\mathbb{R}^2$, define

$$I(\mathcal{L}, P) := \#\{\text{point-line incidences}\} = \#\{(\ell, p) \in \mathcal{L} \times P \mid p \in \ell\}$$

It turns out that we can say how large this can be, up to a constant.

Theorem 7.2.1 (Szemerédi–Trotter). $I(\mathcal{L}, P) = O\left(|\mathcal{L}|^{\frac{2}{3}} |P|^{\frac{2}{3}} + |\mathcal{L}| + |P|\right)$

Proof. Given an $\mathcal{L}, P$ system, consider the graph drawing D whose vertices are the points in P and whose edges are intervals of lines in $\mathcal{L}$ with endpoints in P, with no interior points in P.

We first assume that every line of $\mathcal{L}$ contains at least 2 points of P. In the general case, we remove the lines containing fewer than 2 points of P, get our upper-bound for the lines that remain, and then add them back, which adds at most $|\mathcal{L}|$ incidences; these are absorbed into the $|\mathcal{L}|$ term of the bound.

Let $n = |P|$ be the number of vertices. How many edges do I have? I should have

$$m = I(\mathcal{L}, P) - |\mathcal{L}| = \sum_{\ell \in \mathcal{L}} (\#\{p \in P \mid p \in \ell\} - 1)$$

If we have $4n \leq m$, ie,

$$4|P| \leq I(\mathcal{L}, P) - |\mathcal{L}|$$

then we can apply Theorem 7.1.10 to conclude that

$$\mathrm{cr}(D) \geq \frac{(I(\mathcal{L}, P) - |\mathcal{L}|)^3}{64|P|^2}$$

sorry □

7.2.2 Sums and Products

Fix some $A \subseteq \mathbb{Z}$. Define

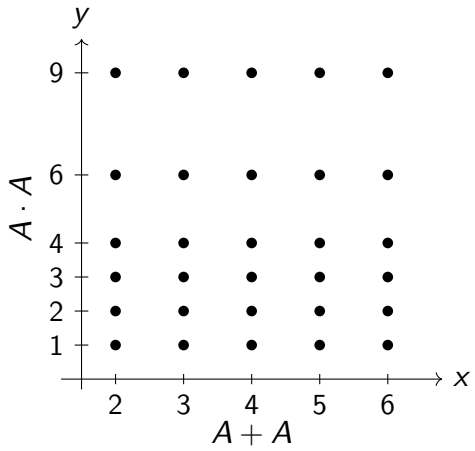
$$A + A := \{a + b \mid a, b \in A\}$$
$$A \cdot A := \{a \cdot b \mid a, b \in A\}$$

Is there a set A for which both $A + A$ and $A \cdot A$ are small?

To answer this, we turn it into an incidence problem. Define

$$P := (A + A) \times (A \cdot A) = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x \in A + A,\ y \in A \cdot A\}$$

For example, when $A = \{1, 2, 3\}$, we have $A + A = \{2, \ldots, 6\}$ and $A \cdot A = \{1, 2, 3, 4, 6, 9\}$, and P looks like this.



The incidence bound answers our question: $A + A$ and $A \cdot A$ cannot both be small.

Theorem 7.2.2. *There is a constant $c > 0$ such that for every finite $A \subseteq \mathbb{Z}$,*

$$|A + A| \, |A \cdot A| \geq c \, |A|^{\frac{5}{2}}$$

In particular, $\max\{|A + A|, |A \cdot A|\} \geq \sqrt{c} \, |A|^{\frac{5}{4}}$.

Proof. `sorry` □

7.3 The Polynomial Method

The rest of this chapter is about lines again: lines in $\mathbb{R}^3$ and the points where three of them meet, and then lines over a finite field. The tool for all of it is polynomials, so let's now take a detour into the world of polynomials.

The arguments that follow all run the same way. We argue by contradiction, and fix a nonzero polynomial p that vanishes on the set we care about, of degree as small as Lemma 7.3.1 allows; if the set is small, so is this degree. We then show that p vanishes at many more points than we asked it to, usually by restricting it to a line: there it is a polynomial in one variable of degree at most $\deg(p)$, so if it vanishes at more than $\deg(p)$ points of the line, it vanishes on all of it.

Finally, we show that a polynomial of such small degree that vanishes at so many points must be the zero polynomial, which contradicts our choice of p. This last step is the delicate one, because vanishing everywhere is not enough on its own: over $\mathbb{F}_q$ we need the degree to be at most $q-1$ (Lemma 7.3.3), and over $\mathbb{R}$ we take p of least degree and differentiate.

Our notation for this course isn't amazing. Let $\mathbb{F}$ be a field.

Notation. $\text{poly}_D(\mathbb{F}^n)$ denotes the set

$$\{p \in \mathbb{F}[x_1, \ldots, x_n] \mid \deg_{\text{tot}}(p) \leq D\}$$

where $\deg_{\text{tot}}$ denotes the total degree of the polynomial.

The monomials form a basis of this space. By a stars and bars argument, we can show that the number of monomials is $\binom{D+n}{n}$. So this is the dimension of this space.

7.3.1 Polynomials Vanishing on a Set

Observe the following: if $S \subseteq \mathbb{F}^n$ is such that $|S| < \dim(\text{poly}_D(\mathbb{F}^n))$, there is some nonzero $p \in \text{poly}_D(\mathbb{F}^n)$ that vanishes on S. The proof is simple: consider the evaluation map

$$\text{ev} : \text{poly}_D(\mathbb{F}^n) \to \mathbb{F}^S : p \mapsto (p(s))_{s \in S}$$

Rank-nullity[1] and the inequality $|S| <$ (dimension of space) tells us the dimension of the kernel is positive, which gives us some p as desired.

The degree bound comes straight out of the dimension count.

Lemma 7.3.1. *If $S \subseteq \mathbb{F}^n$ is finite, there is a nonzero polynomial of degree at most $n|S|^{\frac{1}{n}}$ that vanishes on S.*

Proof. Take $D = n|S|^{\frac{1}{n}}$. Then $\dim_{\mathbb{F}}(\text{poly}_D(\mathbb{F}^n)) = \binom{\lfloor D \rfloor + n}{n} > \frac{D^n}{n!} = \frac{n^n}{n!}|S| \geq |S|$, so by rank-nullity, the evaluation map has a nontrivial kernel. □

[1] Tarun would be so proud!

7.3.2 Polynomials Vanishing Everywhere

Recall that a polynomial evaluating to zero on all inputs does not make it the zero polynomial.

Counterexample 7.3.2. The polynomial $x(x-1) \in \mathbb{F}_2[x]$ evaluates to zero at all points in $\mathbb{F}_2$ but is not equal to the zero polynomial.

But it turns out that under some conditions we can make this fact true.

Lemma 7.3.3. *If $p \in \text{poly}_{q-1}(\mathbb{F}_q^n)$ and $p(x) = 0$ for all $x \in \mathbb{F}_q^n$, then p is the zero polynomial.*

Proof. Argue by induction on n. When $n = 1$, the result follows by polynomial division and applying the fact that each single-variable nonzero polynomial of degree d has at most d roots.

For the inductive step, consider a polynomial $p(x) \in \text{poly}_{q-1}(\mathbb{F}_q^n)$. Write

$$p(x_1, \ldots, x_n) = \sum_{j=0}^{q-1} p_j(x_1, x_2, \ldots, x_{n-1})\, x_n^j$$

for some polynomials $p_j \in \mathbb{F}_q[x_1, \ldots, x_{n-1}]$. Fix $(x_1, \ldots, x_{n-1}) \in \mathbb{F}_q^{n-1}$. Then $\sum_j p_j(x_1, \ldots, x_{n-1})\, t^j$ is a single-variable polynomial in t of degree at most $q-1$ that vanishes on all of $\mathbb{F}_q$, so by the single-variable case each $p_j(x_1, \ldots, x_{n-1})$ is zero. So each p_j, which has total degree at most $q-1$, vanishes on all of $\mathbb{F}_q^{n-1}$, and is zero by the induction hypothesis. □

7.3.3 Directional Derivatives

Over $\mathbb{R}$, we can also differentiate. Given $p \in \mathbb{R}[x, y, z]$, consider the gradient

$$\nabla p = \left(\frac{\partial}{\partial x} p, \frac{\partial}{\partial y} p, \frac{\partial}{\partial z} p \right)$$

At any point $b = (b_0, b_1, b_2)$, and for any unit vector a, the directional derivative of p at b in the direction of a is given by $\langle \nabla p(b), a \rangle$.

7.4 Lines in $\mathbb{R}^3$

We now put the polynomial method of Section 7.3 to work, starting with lines in $\mathbb{R}^3$.

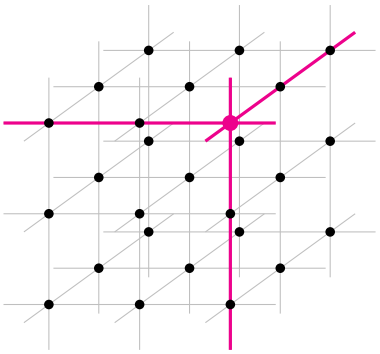
Definition 7.4.1 (Joint). A **joint** in $\mathbb{R}^3$ is an intersection point of 3 non-coplanar lines.

In other words, any point at which three lines that do not share a plane in common meet is called a joint.

We ask ourselves the following question: how many joints can a collection of L lines have?

Let's denote the greatest such number $J(L)$.

Here's one construction: just an $n \times n \times n$ grid. This gives you $3n^2$ lines and n^3 joints.



The case $n = 3$ is drawn above, with the three lines through one of the joints highlighted.

To bound $J(L)$ from above, we show that some line is always in few joints.

Lemma 7.4.2. *In any collection of L lines, some line participates in $\leq 3J^{\frac{1}{3}}$ joints.*

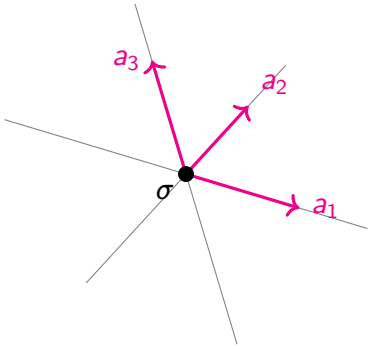
Proof. Suppose not. Ie, suppose we have a collection of L lines in which every line participates in $> 3J^{\frac{1}{3}}$ joints.

Consider a nonzero polynomial p over $\mathbb{R}$ in 3 variables of the least degree $D \leq 3J^{\frac{1}{3}}$ that vanishes on the joints. Such a polynomial exists by Lemma 7.3.1.

We've assumed every line in our collection participates in $> 3J^{\frac{1}{3}}$ joints. Then for any line ℓ of the

form $at + b$ with $a, b \in \mathbb{R}^3$, $p(at + b)$ is a single variable polynomial of degree at most $D \leq 3J^{\frac{1}{3}}$, vanishing at $> D$ points. So p is zero on ℓ. In other words, p is zero on all lines in our collection.

Given a joint σ there are three directions a_1, a_2, a_3 of lines through σ. There may well be more than three lines of our collection through σ, but by definition three of them are not coplanar, and those are the three we take. These three directions form a basis.



The directional derivatives of p at σ in all these directions a_1, a_2, a_3 must be zero. Indeed, p vanishes on the line $\{a_i t + \sigma \mid t \in \mathbb{R}\}$, so $t \mapsto p(a_i t + \sigma)$ is the zero polynomial, and its derivative at $t = 0$, which by the chain rule is $\langle \nabla p(\sigma), a_i \rangle$, is zero. So $\nabla p(\sigma)$ must itself be zero, because its inner product with all elements of a basis is zero. In particular, all three partial derivatives also vanish at every joint. But this necessitates that either one of these is a nonzero polynomial of smaller degree than p or p is the zero polynomial. Either way, we have a contradiction. □

Using the lemma, we notice that

$$\begin{aligned} J(L) &\leq J(L-1) + 3J(L)^{\frac{1}{3}} \\ &\leq J(L-2) + 3 \cdot 2 \cdot J(L)^{\frac{1}{3}} \\ &\leq J(L-3) \cdots \\ &\leq 3 \cdot L \cdot J(L)^{\frac{1}{3}} \end{aligned}$$

The first inequality comes from taking L lines with $J(L)$ joints and deleting a line that, by the lemma, is in at most $3J(L)^{\frac{1}{3}}$ of them; every other joint survives. Repeating this, and using $J(L-1) \leq J(L)$, each of the L deletions costs at most $3J(L)^{\frac{1}{3}}$ joints, which is the last line. Rearranging gives $J(L) \leq (3L)^{\frac{3}{2}}$.

7.5 Nikodym and Kakeya Sets

We now leave $\mathbb{R}^3$ for $\mathbb{F}_q^n$, where a line is a set $\{at + b \mid t \in \mathbb{F}_q\}$ with $a, b \in \mathbb{F}_q^n$ and $a \neq 0$, and so has exactly q points. We ask how small a set can be if it contains almost all of a line through every point, or all of a line in every direction.

7.5.1 Nikodym Sets

Definition 7.5.1 (Nikodym Set). A set $N \subseteq \mathbb{F}_q^n$ is **Nikodym** if $\forall x \in \mathbb{F}_q^n$, $\exists L(x)$ a line through x such that all points of $L(x)$ except for x itself are in N.

In $\mathbb{R}^2$ it means the unit square has a subset N of measure zero such that every point x of the square outside N lies on a line whose other points in the square all lie in N. That such a subset exists is a theorem of Nikodym, from 1927, so in $\mathbb{R}^2$ a set of this kind can be negligible. Over $\mathbb{F}_q$ this is far from the case.[2]

Theorem 7.5.2 (Dvir, 2008). *Any Nikodym set in $\mathbb{F}_q^n$ has size at least $\frac{q^n}{(2n)^n}$.*

Proof. Fix a Nikodym set N. Choose a nonzero polynomial p of degree $\leq n|N|^{\frac{1}{n}}$ that vanishes on N. If $|N| < \left(\frac{q}{2n}\right)^n$ then $\deg(p) \leq q - 1$. By Lemma 7.3.3, if $p(x) = 0$ for all $x \in \mathbb{F}_q^n$, then p is the zero polynomial.

As N is Nikodym, for each $x \in \mathbb{F}_q^n$, there is a line $L(x) = at + b$ with all points of $L(x)$ except for x itself lying in N. The restriction of my polynomial to this line is always a single-variable polynomial such that `sorry` □

7.5.2 Kakeya Sets

Asking instead for a whole line in every direction gives the second kind of set.

[2] The method is Dvir's, from his proof of the finite field Kakeya conjecture: Z. Dvir, *On the size of Kakeya sets in finite fields*, J. Amer. Math. Soc. 22 (2009), 1093–1097, posted to the arXiv in March 2008. The bound for Nikodym sets was written down from it within days, by L. Li, *On the size of Nikodym sets in finite fields*, arXiv:0803.3525.

Definition 7.5.3 (Kakeya Set). A set $K \subseteq \mathbb{F}_q^n$ is **Kakeya** if $\forall a \in \mathbb{F}_q^n \setminus \{0\}$, $\exists b \in \mathbb{F}_q^n$ such that

$$\{at + b \mid t \in \mathbb{F}_q\} \subseteq K$$

The same method shows that Kakeya sets are large too.

Theorem 7.5.4. *Any Kakeya set in $\mathbb{F}_q^n$ has size at least $\left(\frac{q}{10n}\right)^n$.*

Proof. Suppose not. Choose a nonzero polynomial p of degree at most $n|K|^{\frac{1}{n}}$ that vanishes on K. Write $c_n = \left(\frac{1}{10n}\right)^n$. If $|K| < c_n q^n$, then $\deg(p) \leq q - 1$.

For all $a \in \mathbb{F}_q^n \setminus \{0\}$, there is some b such that the line

$$L_{a,b} = \{at + b \mid t \in \mathbb{F}_q\}$$

is contained in K. We know, in particular, that

$$\phi(t) = p(at + b)$$

is the zero polynomial.

Let $D = \deg(p)$. Write $p = p_D + R$, where p_D is the degree D part of p (ie, the part with all monomials having degree D) and R is whatever's left. Observe that

$$\phi(t) = p_D(a)t^D + \cdots + \text{something}$$

The fact that $\phi(t)$ vanishes tells us that $p_D(a) = 0$ for all $a \in \mathbb{F}_q^n \setminus \{0\}$. But also $p_D(0) = 0$. So in fact $p_D(a) = 0$ for all $a \in \mathbb{F}_q^n$. So by Lemma 7.3.3, $p_D = 0$, a contradiction. □

So Lemma 7.3.3 really is quite a useful tool! Let's now explore more results of this flavour.

Chapter 8

Zeros of Polynomials

We now take the polynomials of Section 7.3 away from geometry. The first half of the chapter asks when a polynomial has to be nonzero somewhere on a grid $S_1 \times \cdots \times S_n$, and uses the answer to add sets in $\mathbb{F}_p$ and to colour graphs from lists. The second half works over $\mathbb{F}_q$, where summing a polynomial over every point counts its zeros modulo the characteristic.

8.1 Polynomials on Grids

A nonzero polynomial in one variable of degree d has at most d roots. This section is about what that fact becomes in several variables, where a nonzero polynomial can vanish at infinitely many points, but not at too many points of a grid.

8.1.1 The Schwartz–Zippel Lemma

The first answer is probabilistic: a nonzero polynomial is nonzero at most points of a grid.

Lemma 8.1.1 (Schwartz-Zippel). *Let $\mathbb{F}$ be an arbitrary field and let $p \in \mathrm{poly}_d(\mathbb{F}^n)$ be nonzero. For any finite subset $S \subset \mathbb{F}$, if $s_1, \ldots, s_n \in S$ are chosen uniformly at random (with replacement), then*

$$\Pr(p(s_1, \ldots, s_n) = 0) \leq \frac{d}{|S|}$$

This essentially tells us that a nonzero polynomial of degree d cannot vanish on more than a $\frac{d}{|S|}$ fraction of any grid-set of the form

$$\underbrace{S \times \cdots \times S}_{n \text{ times}}$$

Let's think about how we'd prove Schwartz-Zippel. Suppose we choose $s_1, \ldots, s_n$ randomly. View $p(x_1, \ldots, x_n)$ as a single-variable polynomial in x_n, ie, as an element of $\mathbb{F}[x_1, \ldots, x_{n-1}][x_n]$. Write

$$p(s_1, \ldots, s_n) = p_0(s_1, \ldots, s_{n-1}) + p_1(s_1, \ldots, s_{n-1})\, s_n + p_2(s_1, \ldots, s_{n-1})\, s_n^2 + \cdots + p_k(s_1, \ldots, s_{n-1})\, s_n^k$$

We might want to argue that there are at most d roots of this one-variable polynomial in s_n or something, but the problem is that this does not address the case where this polynomial is 0.

Proof of Lemma 8.1.1 (Schwartz-Zippel). Argue by induction on the number of variables n. The base case $n = 1$ is clear.

Let k be the largest degree of x_n in p. Ie, choose k so that

$$p(x_1, \ldots, x_n) = x_n^k q(x_1, \ldots, x_{n-1}) + r(x_1, \ldots, x_n)$$

where r has degree less than k in x_n. We choose $x_1 = s_1, x_2 = s_2, \ldots, x_{n-1} = s_{n-1}$ first, and $x_n = s_n$ last. By induction, $\Pr(q = 0) \leq \frac{d}{|S|}$; in fact, as $x_n^k q$ is the part of p of highest degree in x_n, q has degree at most $d - k$, so induction gives $\Pr(q = 0) \leq \frac{d-k}{|S|}$. If $q(s_1, \ldots, s_{n-1}) \neq 0$, then $p(s_1, \ldots, s_{n-1}, x_n)$ is a nonzero polynomial in x_n alone, of degree k, so it has at most k roots, and the choice of $x_n = s_n$ hits one of them with probability at most $\frac{k}{|S|}$. So

$$\Pr(p = 0) \leq \Pr(q = 0) + \Pr(p = 0 \text{ and } q \neq 0) \leq \frac{d-k}{|S|} + \frac{k}{|S|} = \frac{d}{|S|}$$

□

8.1.2 The Combinatorial Nullstellensatz

If we know that one of the top-degree coefficients of p is nonzero, we can say more: p is nonzero somewhere on a grid whose sides are only just longer than the exponents in that monomial.

Theorem 8.1.2 (Combinatorial Nullstellensatz). *Suppose $p \in \mathrm{poly}_D(\mathbb{F}^n)$, where $D = j_1 + \cdots + j_n$, is a polynomial with a nonzero coefficient for $x_1^{j_1} x_2^{j_2} \cdots x_n^{j_n}$. Then, for any $S_1, \ldots, S_n \subseteq \mathbb{F}$ with $|S_i| \geq j_i + 1$, we have that some $x \in S_1 \times \cdots \times S_n$ has $p(x) \neq 0$.*

Remark. Note that the condition that the coefficient for $x_1^{j_1} x_2^{j_2} \cdots x_n^{j_n}$ is nonzero implies that the total degree is at least $j_1 + \cdots + j_n$.

Proof. We argue by induction on $D = j_1 + \cdots + j_n$. If $D = 0$, then p is a nonzero constant, and there is nothing to prove. So suppose $D \geq 1$, and that the result holds for all polynomials of smaller degree.

Fix p as above. WLoG, assume $j_1 \geq 1$ (so that $|S_1| \geq 2$). For $s \in \mathbb{F}$, write

$$p(x_1, \ldots, x_n) = (x_1 - s)\, q_s(x_1, \ldots, x_n) + r_s(x_2, \ldots, x_n)$$

Notice that

$$p(s, s_2, s_3, \ldots, s_n) = r_s(s_2, \ldots, s_n)$$

If $p(s, s_2, \ldots, s_n) = 0$ then $r_s(s_2, \ldots, s_n) = 0$.

Now suppose, towards a contradiction, that p vanishes on all of $S_1 \times \cdots \times S_n$, and fix $s \in S_1$. Then r_s vanishes on $S_2 \times \cdots \times S_n$, and since r_s does not involve x_1, it vanishes on all of $S_1 \times \cdots \times S_n$. So $(x_1 - s)\, q_s$ vanishes on $S_1 \times \cdots \times S_n$, and as $x_1 - s \neq 0$ when $x_1 \neq s$, q_s vanishes on $(S_1 \setminus \{s\}) \times S_2 \times \cdots \times S_n$.

We want to apply the induction hypothesis to q_s, so we need to know what it looks like. Write $p = \sum_k c_k x_1^k$, where each $c_k \in \mathbb{F}[x_2, \ldots, x_n]$ has degree at most $D - k$. As $x_1^k - s^k = (x_1 - s)\left(x_1^{k-1} + s x_1^{k-2} + \cdots + s^{k-1}\right)$, we can take

$$q_s = \sum_k c_k \left(x_1^{k-1} + s x_1^{k-2} + \cdots + s^{k-1}\right) \qquad \text{and} \qquad r_s = \sum_k c_k s^k$$

and every monomial of q_s has degree at most $(D - k) + (k - 1) = D - 1$. In particular, q_s has no monomial of degree D, and r_s has no monomial involving x_1. So when we compare the coefficients of $x_1^{j_1} x_2^{j_2} \cdots x_n^{j_n}$ on the two sides of $p = (x_1 - s)\, q_s + r_s$, the only contribution on the right comes from multiplying the $x_1^{j_1 - 1} x_2^{j_2} \cdots x_n^{j_n}$ term of q_s by x_1. Hence $q_s \in \mathrm{poly}_{D-1}(\mathbb{F}^n)$ has a nonzero

coefficient for $x_1^{j_1-1}x_2^{j_2}\cdots x_n^{j_n}$. Since $|S_1 \setminus \{s\}| \geq j_1 = (j_1 - 1) + 1$, the induction hypothesis gives a point of $(S_1 \setminus \{s\}) \times S_2 \times \cdots \times S_n$ at which q_s is nonzero, which is a contradiction. □

8.1.3 The Cauchy–Davenport Theorem

Our first application is to sums of sets in $\mathbb{F}_p$, where $A + B = \{a + b \mid a \in A, b \in B\}$.

Corollary 8.1.3 (Cauchy-Davenport). *For all nonempty $A, B \subseteq \mathbb{F}_p$,*

$$|A + B| \geq \min(p, |A| + |B| - 1)$$

Proof. This is "trivial" if $|A| + |B| > p$. Indeed, for any $x \in \mathbb{F}_p$, the sets A and $x - B = \{x - b \mid b \in B\}$ have sizes adding up to more than p, so by the pigeonhole principle they share some element $a = x - b$, and then $x = a + b \in A + B$. So $A + B = \mathbb{F}_p$. So let's assume that $|A| + |B| \leq p$.

Assume, towards contradiction, that $A + B \subseteq C$, where $|C| = |A| + |B| - 2$. Define

$$p(x, y) = \prod_{c \in C} (x + y - c)$$

Notice that $p(x, y) = 0$ for all $x \in A$ and $y \in B$. Moreover, notice that $x^{|A|-1}y^{|B|-1}$ has coefficient $\binom{|A|+|B|-2}{|A|-1}$. sorry □

8.1.4 List Colourings

The second is to colouring graphs (cf. Definition 2.2.1), where each vertex now comes with its own list of allowed colours.

Definition 8.1.4 (List Chromatic Number). For a graph G, define the **list chromatic number** $\chi'(G)$ to be the smallest k such that for any assignment of lists of size k to vertices of G, there is a proper colouring from the list.

Our strategy for finding upper-bounds will be to apply the Combinatorial Nullstellensatz to the

following polynomial:

$$p(x_1, \ldots, x_n) = \prod_{i \sim j} (x_i - x_j)$$

where we label the vertices of the graph $\{1, \ldots, n\}$ and take the product over adjacent vertices, ie, over edges.

The field over which we define this polynomial does not matter much, so long as it has enough elements. The colours in our lists have to live in the field, but whether a colouring is proper only depends on which vertices get the same colour, so we can relabel the colours as distinct elements of any field with at least as many elements as there are colours in all the lists together. So we can always get away with taking the field to be $\mathbb{Q}$ or $\mathbb{R}$. In fact, $\mathbb{Q}$ is never a worse choice than $\mathbb{F}_q$: the coefficients of p are integers, and the Combinatorial Nullstellensatz asks for one of them to be nonzero, which can fail in $\mathbb{F}_q$ for an integer that is nonzero in $\mathbb{Q}$.

8.2 Polynomials over Finite Fields

Throughout this section, q is a prime power and $\mathbb{F}_q$ is the field with q elements.

8.2.1 Sums over $\mathbb{F}_q$

Summing a power of x over all of $\mathbb{F}_q$ gives zero, unless the power is large.

Theorem 8.2.1. *For all $k < q - 1$,*

$$\sum_{x \in \mathbb{F}_q} x^k = 0$$

Proof. For $k = 0$ the sum is $q = 0$. Otherwise, note that $x^k - 1$ has at most $k < q - 1$ roots over $\mathbb{F}_q$. So there is certainly some $a \in \mathbb{F}_q \setminus \{0\}$ that is *not* a root of this polynomial. So $a^k \neq 1$. Since $x \mapsto ax$ is a bijection,

$$\sum_{x \in \mathbb{F}_q} x^k = \sum_{x \in \mathbb{F}_q} (ax)^k = a^k \sum_{x \in \mathbb{F}_q} x^k$$

As $a^k \neq 1$, the only way this equality can hold is if $\sum_{x \in \mathbb{F}_q} x^k = 0$. □

Summing over $\mathbb{F}_q^n$ instead, the same is true of any polynomial of small enough degree.

Theorem 8.2.2. *Let $p \in \mathbb{F}_q[x_1, \ldots, x_n]$. Assume $\deg(p) < n(q-1)$. Then*

$$\sum_{\mathbf{x} \in \mathbb{F}_q^n} p(\mathbf{x}) = 0$$

Proof. Enough to do the case where $p(\mathbf{x})$ is a monomial. `sorry` □

8.2.2 Fermat's Little Theorem

We'll now prove Fermat's Little Theorem combinatorially.

Theorem 8.2.3 (Fermat's Little Theorem)**.** *For p a prime, for all $a \in \mathbb{Z}$, $a^p \equiv a \pmod{p}$. Equivalently, for p a prime, for all $a \in \mathbb{Z}$ such that $p \nmid a$, $a^{p-1} \equiv 1 \pmod{p}$.*

Proof. For $p \nmid a$, we know that $x \mapsto ax$ is a bijection of $\mathbb{F}_p \setminus \{0\}$. So

$$\prod_{x \in \mathbb{F}_p \setminus \{0\}} x = \prod_{x \in \mathbb{F}_p \setminus \{0\}} (ax) = a^{p-1} \prod_{x \in \mathbb{F}_p \setminus \{0\}} x$$

in $\mathbb{F}_p$, and cancelling the product, which is nonzero, gives $a^{p-1} = 1$. □

Of course the corresponding result holds in $\mathbb{F}_q$ as well.

Theorem 8.2.4. *For all $a \in \mathbb{F}_q \setminus \{0\}$, $a^{q-1} = 1$.*

"Proof". $\mathbb{F}_q$ is, by definition, the splitting field of $x^q - x$ over $\mathbb{F}_p$, where q is a power of p. So when $a \in \mathbb{F}_q$ is nonzero,

$$a^q - a = 0 \iff a\left(a^{q-1} - 1\right) = 0 \iff a^{q-1} - 1 = 0$$

□

8.2.3 The Chevalley–Warning Theorem

Let's now study systems of polynomials over $\mathbb{F}_q$.

Theorem 8.2.5 (Chevalley-Warning). *Fix $p_1, \ldots, p_k \in \mathbb{F}_q[x_1, \ldots, x_n]$ with $\sum_{i=1}^{k} \deg(p_i) < n$. Let p be the characteristic of $\mathbb{F}_q$. Then*

$$p \mid \#\left\{\mathbf{x} \in \mathbb{F}_q^n \;\middle|\; p_1(\mathbf{x}) = p_2(\mathbf{x}) = \cdots = p_k(\mathbf{x}) = 0\right\}$$

Proof. Define the polynomial

$$Q(\mathbf{x}) = \prod_{i=1}^{k} \left(1 - p_i(\mathbf{x})^{q-1}\right)$$

What's interesting about the way we've defined Q is that

- $Q(\mathbf{x}) = 0$ iff $p_i(\mathbf{x}) \neq 0$ for *some* i
- $Q(\mathbf{x}) = 1$ iff $p_i(\mathbf{x}) = 0$ for *every* i

Consider the quantity

$$\mathbb{F}_q \ni z = \sum_{\mathbf{x} \in \mathbb{F}_q^n} Q(\mathbf{x})$$

Notice that $\deg(Q) < n(q-1)$, so by Theorem 8.2.2, $z = 0$. We can write

$$z = \underbrace{1 + \cdots + 1}_{N \text{ times}}$$

where $N = \#\left\{\mathbf{x} \in \mathbb{F}_q^n \;\middle|\; p_1(\mathbf{x}) = p_2(\mathbf{x}) = \cdots = p_k(\mathbf{x}) = 0\right\}$. As $N = 0$ in $\mathbb{F}_q$, we must have $p \mid N$. □

8.2.4 The Erdős–Ginzburg–Ziv Theorem

Chevalley–Warning has a striking consequence for the integers.

Theorem 8.2.6 (Erdős-Ginzburg-Ziv). *Out of any $2n-1$ integers, there are n whose average is an integer. More precisely, for all $a_1, \ldots, a_{2n-1} \in \mathbb{Z}$, $\exists I \subseteq \{1, \ldots, 2n-1\}$ such that $|I| = n$ and $\sum_{i \in I} a_i \equiv 0 \pmod{n}$.*

Proof. Consider the case $n = p$, with p a prime. Notice that if $\mathbf{x} = (x_1, \ldots, x_{2p-1}) \in \mathbb{F}_p^{2p-1}$,

$$\sum_{i=1}^{2p-1} x_i^{p-1} = \sum_{i=1}^{2p-1} \mathbf{1}_{x_i \neq 0}$$

Fix $a_1, \ldots, a_{2p-1} \in \mathbb{Z}$, and consider the polynomials

$$p_1(\mathbf{x}) = \sum_{i=1}^{2p-1} a_i x_i^{p-1} \qquad \text{and} \qquad p_2(\mathbf{x}) = \sum_{i=1}^{2p-1} x_i^{p-1}$$

in $\mathbb{F}_p[x_1, \ldots, x_{2p-1}]$, reading the a_i mod p. Their degrees add up to at most $2p-2 < 2p-1$, so by Chevalley–Warning, the number of common zeros of p_1 and p_2 is a multiple of p. One of them is $\mathbf{x} = 0$, so there is another, $\mathbf{x} \neq 0$. Let $I = \{i \mid x_i \neq 0\}$, which is nonempty. By Fermat's Little Theorem, x_i^{p-1} is 1 if $i \in I$ and 0 otherwise, so in $\mathbb{F}_p$,

$$0 = p_2(\mathbf{x}) = |I| \qquad \text{and} \qquad 0 = p_1(\mathbf{x}) = \sum_{i \in I} a_i$$

So p divides $|I|$, and as $1 \leq |I| \leq 2p-1$, $|I| = p$. And $\sum_{i \in I} a_i \equiv 0 \pmod{p}$.

This takes care of the prime case.

Now, assume the result is proven for $n = a$ and $n = b$. We prove the result for $n = ab$.

Fix $c_1, c_2, \ldots, c_{2ab-1} \in \mathbb{Z}$. I repeatedly find and remove b terms summing to 0 mod b. I can do this $2a-1$ times, to get disjoint sets $B_1, B_2, \ldots, B_{2a-1}$, each summing as $\sum_{c \in B_i} c = y_i \equiv 0 \pmod{b}$.

Using the result for a on the integers $\frac{y_1}{b}, \ldots, \frac{y_{2a-1}}{b}$, I can find a of the sets $B_{i_1}, B_{i_2}, \ldots, B_{i_a}$ such that $\sum_{j=1}^{a} \frac{y_{i_j}}{b} \equiv 0 \pmod{a}$, so that $\sum_{j=1}^{a} y_{i_j} \equiv 0 \pmod{ab}$. But this is the sum of the ab numbers in $B_{i_1} \cup \cdots \cup B_{i_a}$, which is what we wanted. □

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